

Synergistic Authentication in Beam Alignment for Secure mmWave Communications

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Abstract—The efficacy of millimeter wave (mmWave) multiple-input multiple-output (MIMO) systems relies on a precise beam alignment protocol to establish high-gain directional links. However, the super-directive directional beams and the broadcast nature of training signals during the alignment phase render communication devices extremely susceptible to identity-based impersonation attacks. To combat such security threats, two-modal features including antenna array-specific mutual coupling (MC) and spatial angle of arrival (AoA) are jointly exploited to design two hybrid physical (PHY)-layer authentication schemes including a linear weighted combination (LWC) scheme and a two-threshold combined one. Moreover, analytical expressions for authentication performance metrics like detection and false alarm probabilities are also derived to conduct a theoretical analysis under both Gaussian and Laplace AoA scenarios. To fully achieve the collaborative authentication capability of the LWC scheme, a sum-weighted approximation technique is utilized to establish the mapping relationship between feature weights and performance metrics. Based on such a relationship, an optimization problem for optimal feature weight values is then formulated to maximize the detection accuracy under a false alarm probability constraint. Then, Cramér-Rao lower bounds are derived to examine the theoretical performance bound of the developed authentication methods. Finally, simulation results are provided to validate the proposed theoretical models and demonstrate the effectiveness of the designed schemes against impersonation attacks.

Index Terms—Identity authentication, antenna array fingerprint, wireless security, impersonation attacks.

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I. INTRODUCTION

A. Background and Motivation

MILLIMETER wave (mmWave) communication can facilitate high-speed data transfer and improved spectrum efficiency and thus is envisioned as a crucial candidate for the future beyond fifth-generation (5G) wireless communication networks [1]. In addition, the integrated mmWave multiple-input multiple-output (MIMO) system serves as the key enabler to support crucial applications with seamless wireless connectivity and ultra-low latency requirements, such as autonomous driving, smart cities, and satellite communications [2]. However, the inherent characteristics of mmWave MIMO systems like the broadcast nature of training sequences and super-directive directional transmission lead to exploitable vulnerabilities for identity-based impersonation attacks. In particular, similar to traditional narrow band systems, the inherent openness of the wireless medium grants attackers opportunities to eavesdrop on communication processes for obtaining legitimate user identity information (e.g., media access control address) and signaling formats [3]. Such confidential intelligence significantly aids attackers in forging identities and subsequently impersonating legitimate users through the injection of malicious signals. More importantly, the highly directional transmissions between legitimate transceivers enable the mmWave systems to be more susceptible to identity impersonation attacks. The attacker can potentially intercept legitimate communication operations and transmit deceptive beam signals along the intended communication path, thus misleading the receiver into aligning with a malicious source and achieving unauthorized access to critical network resources [4]. Once the impersonation attack is completed, more serious damage such as data tampering and malicious command injection will threaten the concerned system [5]. Therefore, designing a reliable authentication scheme to combat impersonation attacks is of vital significance for securing the mmWave MIMO systems.

Authentication is one of the most imperative security services utilized to confirm the identity authority of the terminal devices/users accessing the network to prevent severe security threats such as impersonation attacks and unauthorized access [6]. Due to the prominent advantages in terms of alleviated computational burden, lower time overhead and improved network compatibility, physical (PHY)-layer authentication has garnered considerable attention as a promising lightweight authentication method [7]. PHY-layer authentication generally exploits the unique and intrinsic physical characteristics of device hardware imperfections and/or wireless channels to validate the identities of devices/users, which is roughly classified

into three categories: 1) hardware impairment-based authentication [8], [9], [10], [11], [12]; 2) channel-based authentication [13], [14], [15], [16], [17], [18]; and 3) multi-feature authentication [19], [20], [21], [22], [23] (Please refer to Section II for detailed related works).

Recently, some preliminary studies have been devoted to the deployment of PHY-layer authentication into the mmWave MIMO system [24], [25], [26], [27], [28]. For PHY-layer authentication based on hardware impairments, the authors in [24] exploit the signal-to-noise ratio (SNR) trace generated from the sector-level sweep procedure to distinguish 60 GHz IEEE 802.11ad devices. For the channel-based authentication, Afeef et al. leverage the prominent component of mmWave MIMO channel in the beamspace domain to construct a distance-based signature for combating the spoofing attacks [26]. Also, we capture the mmWave channel sparsity to effectively differentiate between legal and illegal drones [27] for resisting impersonation attacks. For the hybrid authentication, Liu et al. jointly employ the device-specific phase noise and angle domain channel features (including azimuth angle of arrival (AoA), elevation AoA and channel gain) to design a four-dimensional PHY-layer authentication method [28].

B. Limitations of Prior Art

Despite recent progress in designing PHY-layer authentication for mmWave systems, some critical issues are still not well addressed yet owing to the following limitations. First, existing schemes in [24], [25], and [26] independently rely on either a single hardware feature or a single channel one, possessing limited feature space resolution and thus making it difficult to satisfy the secure authentication demands of massive access devices in mmWave networks. Second, the scheme in [28] only considers the Gaussian AoA model and will exhibit degraded performance in the 60 GHz frequency band. This is because the Gaussian assumption inadequately characterizes the true AoA statistical behavior, which often exhibits non-Gaussian characteristics (e.g., a more pronounced peak and heavier tails) in the specific band, thereby leading to suboptimal authentication performance due to the statistical model mismatch. Moreover, the model mismatch enables existing Gaussian model-based theoretical analysis to become inaccurate when applied in the 60 GHz band. This indicates a necessity for a new theoretical framework for the concerned new scenario to fill the analysis gap. Third, while some works in [20], [28] have explored the convenient combination authentication mechanism under the Gaussian model, they merely utilize multiple features to individually validate device authority and then simply combine these decision results to make a final identity judgment. This method fails to fully exploit the synergy between different features, inevitably compromising the overall authentication accuracy.

C. Our Solutions and Key Contributions

To address these limitations, two promising perspectives can be considered to enhance the PHY-layer authentication performance in mmWave MIMO systems. On one hand, the rich and intrinsic hardware fingerprints involved in the massive antenna arrays (e.g., mutual coupling (MC) effect) of mmWave MIMO systems have not been effectively exploited

to enhance the PHY-layer authentication performance. Recent studies in [29], [30] have shown that electromagnetic coupling between antenna elements (also namely MC effect) is regarded as one of the antenna array impairments induced by manufacturing tolerances and environmental factors, which is highly governed by the antenna array and possesses better environmental robustness. This provides a novel insight for device identity authentication in an mmWave MIMO system. On the other hand, AoA exhibits remarkable feature stability and intrinsically involves richer fine-grained spatial information of the device compared to transient channel characteristics such as channel gain and channel phase [31], [32]. Meanwhile, we note that the Laplace distribution model can more accurately capture the actual AoA statistical characteristics in the 60 GHz frequency band [33], and thus we consider it for the scheme design.

Based on the above observations, our preliminary conference work utilized the basic joint MC-AoA authentication under ideal Gaussian assumptions. Building upon this foundation, this paper makes substantial extensions. The primary incremental advances encompass a new Laplace-AoA statistical model and corresponding two threshold combination scheme (TTC) tailored for the 60 GHz band, along with the rigorous derivation of theoretical performance bounds via the Cramér-Rao Lower Bound (CRLB). Integrating these advances, the main contributions of this paper are summarized as follows:

- For the traditional Gaussian AoA scenario, we exploit the antenna array MC and spatial AoA features to propose a novel LWC authentication scheme to resist impersonation and co-located attacks in mmWave MIMO systems.
- We exploit a new Laplace model to accurately capture the statistical behavior of AoA feature in the 60 GHz frequency band and utilize the Laplace scale parameter to effectively depict the spatial characteristics of the AoA. Based on this model, we then design a novel TTC authentication scheme to satisfy various security requirements in mmWave systems.
- We develop a complete theoretical performance analysis framework that addresses the limitations of prior Gaussian model-based analyses. This framework is capable of characterizing authentication performance under both Gaussian and Laplace AoA scenarios by deriving their respective analytical expressions for detection and false alarm probabilities. Further, the CRLB is also derived to evaluate the theoretically optimal authentication performance bound.
- To optimize the authentication performance of the designed LWC scheme, we first bridge a relationship between feature weights and performance metrics by using a sum-weighted approximation technique. Based on the developed relationship, an optimization problem is then formulated to determine the optimal feature weights for maximizing the identity validation accuracy.

The remainder of this paper is organized as follows. Section II summarizes related works. Section III introduces the problem statement and system model. We illustrate the authentication scheme in Section IV and provide theoretical analysis in Section V. The performance evaluation is presented in Section VI and concluding remarks are presented in Section VII. The main notations are summarized in Table I.

TABLE I
SUMMARY OF KEY NOTATIONS AND SYMBOLS

Symbol	Description
N_t, N_r	Numbers of transmit and receive antennas.
L	Number of dominant propagation paths.
P	Number of non-zero MC coefficients.
$\mathbf{H}$	mmWave channel matrix.
α_l	Complex gain of the l -th path.
θ_l, ϕ_l	AoA and AoD of the l -th path.
$\mathbf{a}_r(\theta_l), \mathbf{a}_t(\phi_l)$	Receive/transmit steering vectors.
$\mathbf{C}_r, \mathbf{C}_t$	Receive/transmit MC matrices.
$\mathbf{c}_r, \mathbf{c}_t$	Receive/transmit MC coefficient vectors.
$\mathbf{C}$	Equivalent MC matrix, $\mathbf{C} = \mathbf{C}_r \otimes \mathbf{C}_t$.
$\mathbf{A}$	Ideal steering matrix.
$\mathbf{y}(n, k), \mathbf{v}(n, k)$	Received signal and noise vectors.
N	Number of snapshots.
$\hat{\mathbf{R}}$	Estimated covariance matrix
$\hat{\theta}_l, \hat{\phi}_l, \hat{\mathbf{c}}_t$	Estimated AoA, AoD, and MC feature.
w_θ, w_c	AoA and MC estimation errors.
$\Lambda(\cdot)$	Likelihood ratio test operator.
$\sigma_{w,\theta}^2, \sigma_{w,c}^2$	Variances of estimation errors.
$\mathcal{H}_0, \mathcal{H}_1$	Hypotheses corresponding to Alice and Eve.
$\tau_\theta, \tau_\theta, \tau_c, \tau_G$	Decision thresholds.
ω_1, ω_2	AoA and MC weights in the LWC scheme.
P_f, P_d	False alarm and detection probabilities.
Γ	Target false alarm constraint.
$\text{CRB}(\theta), \text{CRB}(\mathbf{c}_t)$	CRLBs of AoA and MC estimation.
κ_θ, κ_c	AoA and MC fingerprint similarity factors.
δ, N_{grid}	Grid resolution and total search points.

II. RELATED WORK

Hardware Impairment-based Approaches: The hardware impairment-based approaches utilize hardware impairments resulting from the manufacturing process to characterize device identities. Specifically, the authors in [34] leverage the carrier frequency offset (CFO) as an enduring device fingerprint to bolster cryptographic authentication in WLANs. However, such one-dimensional hardware feature possesses limited identity discrimination capability in front of the massive devices in mmWave MIMO systems. To this end, Liu et al. utilize the non-linear phase error stemming from I/Q imbalance and oscillator mismatch to detect rogue WiFi connections with improved 96% detection accuracy [10]. Moreover, a data-independent function model is proposed in [11] to characterize the combined RF chain imperfections from modulation error to power amplification for identification of 33 IoT devices.

Channel-based Approaches: The principle of channel-based approaches mainly resorts to location-specific information such as channel state information (CSI) [13], channel environmental noise [14], channel phase response [15], and channel polarization response [16] to verify the legitimacy of transmitters. Recently, AoA has emerged as a promising spatial feature for channel-based PHY-layer authentication due to its inherent link to the physical location of transmitters. Several studies have explored its viability and limitations from several perspectives [17], [18], [35], [36]. For instance, Pham et al. present the fundamental robustness of AoA by demonstrating that a successful impersonation attack requires an adversary to meet highly stringent location and hardware conditions [17]. However, this robustness can be challenged by more sophisticated attacks. The authors in [18] investigate the vulnerability of AoA-based authentication in analog array systems and show that an attacker with knowledge of the receiver's combiners can mount a successful impersonation attack by falsifying the

estimated AoA. To enhance distinctiveness against such threats, other works propose combining AoA with additional features. Xu et al. introduce a novel scheme that integrates AoA with the rotational state information of the UE to create a unique decision metric for authenticating a base station (BS) [35]. Furthermore, the application of the AoA feature has been extended to specific domains, where Tan et al. propose an AoA-based authentication scheme to effectively address GNSS spoofing attacks in scenarios where the authentic signal is blocked [36].

Multi-feature Approaches: The multi-feature approaches jointly combine various physical features to achieve multiple identity validation. For example, Zhang et al. first utilize the error vector magnitude to model hardware impairment and then combine it with channel gain (CG) to design an authentication scheme in massive MIMO systems [19]. Following this line, they further propose to exploit the CG and phase noise (PN) to jointly design a two-dimensional device identity authentication method [20]. Illi et al. design an improved hybrid scheme by using the CSI and CFO [21]. A recent work in [22] combines the channel response and secret key-generated tag features to propose a challenge-response authentication scheme. Recently, an adaptive PHY-layer authentication scheme leveraging MIMO antenna diversity combines signal magnitudes with real and imaginary components via support vector machines [37]. Moreover, radiation pattern features encompassing gain, phase, and position errors have been jointly exploited for device authentication [38], while a multi-observation multi-channel-attribute architecture has been developed to enhance security in industrial wireless edge networks [39].

Despite these advances, existing multi-feature methodologies possess critical limitations that hinder true synergistic authentication in practical mmWave MIMO scenarios. First, regarding feature vulnerability and scarcity, channel-only schemes remain vulnerable to spatial spoofing (e.g., co-located attacks). While incorporating hardware fingerprints improves robustness, traditional RF features suffer from severe scarcity due to the limited number of RF chains in cost-constrained mmWave systems, compromising overall feature dimensionality. Second, regarding decision synergy, although recent efforts attempt to employ empirical settings or machine learning algorithms to decide feature weight, these static parameter settings and closed box approaches fail to dynamically adjust to real-time signal quality fluctuations and environmental dynamics. Furthermore, they lack a rigorous mathematical foundation to guarantee the optimality of the assigned weights. Third, regarding model mismatch, current frameworks predominantly rely on simplified Gaussian assumptions, which inadequately capture the true non-Gaussian spatial propagation characteristics in the 60 GHz band, inevitably causing theoretical mismatch and suboptimal accuracy.

In contrast, our scheme achieves cross-domain synergistic authentication driven by an adaptive feature weight mechanism, addressing the above limitations across three aspects. First, regarding specific feature pairing, we combine the spatial AoA with the inherent antenna-specific MC effect. Unlike scarce RF-chain features, this MC fingerprint intrinsically becomes richer as the antenna array scales up to achieve the high-precision beamforming required in mmWave massive MIMO systems. Second, regarding the synergistic decision, we design an adaptive feature weight mechanism using a linear weighted combination scheme. Through rigorous mathematical modeling via a sum-weighted approximation technique, we explicitly reveal the

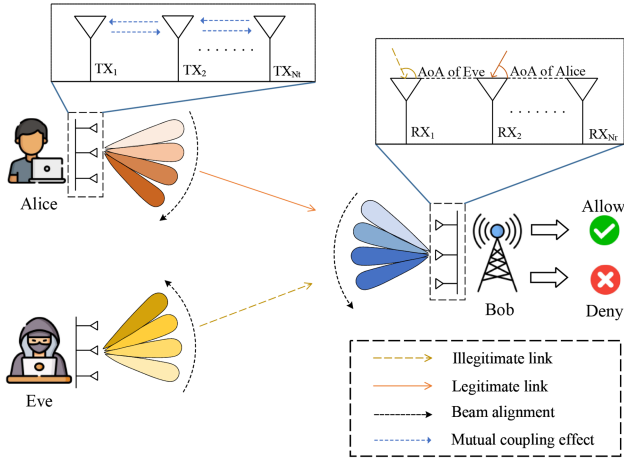


Fig. 1. System model.

relationship between feature weights and authentication performance metrics. An optimization problem is then formulated to obtain the optimal weight allocation, thereby maximizing the identity validation accuracy. Finally, we develop a Laplace AoA analytical framework to accurately characterize practical 60 GHz mmWave spatial propagation and thus extend the applicability of our synergistic scheme to broader mmWave scenarios.

Deep Learning-based Approaches: Recently, deep learning (DL) has emerged as a powerful tool for PHY-layer authentication primarily due to its exceptional automatic feature extraction capabilities. Initial efforts aim to mitigate the impact of time-varying channels on feature stability. Specifically, the authors in [40] propose a single-source domain generalization approach to prevent models from learning channel-induced shortcuts, ensuring the network focuses on intrinsic hardware impairments. In addition, dynamic adaptation methods leveraging cognitive representation learning [41], and schemes utilizing sounding reference signals in 5G NR [42], have been introduced to maintain robust cross-domain identification. Moreover, a bi-interpolative metric learning framework is proposed to accurately identify specific emitters under few-shot conditions by optimizing the feature space distance [43]. More recently, several methods have been explored to alleviate labeling and privacy bottlenecks. For instance, the authors in [44] propose a self-supervised architecture utilizing time-frequency contrastive learning alongside CutMix regularization, significantly reducing the reliance on large-scale annotated datasets. Similarly, federated learning frameworks with feature alignment enable decentralized training without exposing raw signal data [45]. However, despite their competitive accuracy, DL-based schemes generally require extensive training datasets and computational resources, which may be challenging for resource-constrained mmWave devices. Moreover, DL-based models are often characterized as closed box mechanisms, lacking closed-form analytical frameworks to mathematically evaluate security performance bounds.

III. PROBLEM STATEMENT AND SYSTEM MODEL

A. Problem Formulation

We illustrate in Fig. 1 the considered mmWave MIMO communication scenario involving three entities: one BS Bob with

an N_r -element antenna array, a legal user equipment (UE) Alice with an N_t -element antenna array and a potential attacker Eve equipped with N_t antennas. These transceiver parties are assumed to be geographically located in different positions. In particular, Alice resorts to beamforming technology by adjusting the phase and magnitude of antenna array elements to achieve more efficient signal transmission with the intended receiver Bob. Correspondingly, Bob employs beam scanning and alignment to obtain received signals in specified directions and accordingly conducts operations such as information exchange and command execution. The attacker Eve attempts to masquerade as Alice to communicate with Bob for unauthorized access.

It is assumed that the authentication process occurs after an initial communication link has been established. In specific, the authentication process operates under the standard assumption for mmWave MIMO systems that initial beam training and alignment procedures have already been performed, providing a coarse directional estimate for the communication link between the UE (Alice or Eve) and Bob [46], [47]. Then, based on the acquired initial angle range, the subsequent feature extraction process in Section IV provides the fine-grained AoA information for the final identity authentication of the transmitter end. Furthermore, we assume that the device-specific MC model parameters and the spatial AoA are learned based on every legitimate device during a secure offline registration phase [20], [22]. This process allows the receiver to establish unique identity templates, serving as a robust basis for subsequent identity validation.

B. Threat Model

Our threat model assumes a highly capable and active adversary Eve, who seeks unauthorized access by impersonating the legitimate user Alice. To ensure a rigorous evaluation, we explicitly define Eve's prior knowledge acquisition pathways and her advanced adaptive attack execution methods:

- **Knowledge Acquisition via Protocol Vulnerabilities:** Unlike traditional static assumptions, we assume Eve actively exploits the inherent openness of the mmWave beam alignment protocol. By eavesdropping on the unencrypted sector sweep and the subsequent beam report frames [48], Eve can feasibly obtain Alice's pilot sequences and optimal beam ID.
- **Execution of Dynamic Adaptive Attacks:** Based on the acquired knowledge, Eve executes a two-stage impersonation attack. First, Eve obtains the specific model and hardware configuration (e.g., operating center frequencies, communication standards, and nominal array geometry) of the enrolled legitimate devices. She then adopts a commercial off-the-shelf device of the exact same model and configuration to impersonate the legitimate user to communicate with the receiver. Then, to bypass spatial-based authentication, Eve goes beyond traditional co-located attacks (where an attacker relies solely on static physical proximity). Equipped with the cloned hardware and a programmable phased antenna array, Eve dynamically computes and adjusts her beamforming weights to adaptively steer malicious beams along Alice's intended spatial paths [49], [50]. This dynamic beam alignment allows Eve to precisely mimic Alice's AoA fingerprint at Bob's receiver.

C. MmWave Channel Model

Because of the severe signal attenuation and sparse scattering environment involved in the mmWave MIMO system, we exploit the commonly used geometric channel model to characterize the uplink propagation path between the UE and BS [51]. Such uplink mmWave channel $\mathbf{H} \in \mathbb{C}^{N_r \times N_t}$ without antenna impairments is mainly parameterized by the array steering vectors at the transceiver sides and the spatial domain angles. If we use $\mathbf{a}_r(\theta_l) \in \mathbb{C}^{N_r}$ to denote the ideal response vector at the receiver with the AoA parameter θ_l and consider the uniform linear array (ULA) with half-wavelength antenna spacing, then $\mathbf{a}_r(\theta_l)$ can be expressed as

$$\mathbf{a}_r(\theta_l) = \frac{1}{\sqrt{N_r}} [1, e^{j\pi \sin(\theta_l)}, \dots, e^{j\pi(N_r-1) \sin(\theta_l)}]^T. \quad (1)$$

The ideal steering vector $\mathbf{a}_t(\phi_l) \in \mathbb{C}^{N_t}$ with angle of departure ϕ_l can be written in a similar manner. Based on the steering vectors at the transceiver ends, the channel $\mathbf{H}$ with L dominant propagation paths is formulated as

$$\mathbf{H} = \sqrt{\frac{N_r N_t}{L}} \sum_{l=1}^L \alpha_l \mathbf{a}_r(\theta_l) \mathbf{a}_t^H(\phi_l), \quad (2)$$

where α_l is the complex gain of the l -th path.

D. Mutual Coupling Fingerprint Model

1) Mathematical Model of MC Fingerprint: In practice, the actual steering vectors are unavoidably impacted by the antenna calibration errors resulting from gain-phase uncertainty, antenna element location error, and MC effects. Herein, we mainly focus on the MC effects and suppose the negative impacts of the other calibration errors have been perfectly eliminated. The MC effects refer to the electromagnetic coupling between the antenna elements, which are highly determined by the antenna polarization, geometry design, and manufacturing materials [26]. Note that the antenna array geometry specifies the MC structure. Similar to the modeling methods in [52], [53], the MC structure of the ULA can be given by a symmetric Toeplitz matrix. Moreover, the coupling effect is inversely proportional to the distance between two antenna elements, indicating that the MC effect can be considered negligible when the distance is large. If we denote P the number of the non-zero MC coefficients in both transmit and receive arrays, then the MC matrix of the transmit array $\mathbf{C}_t \in \mathbb{C}^{N_t \times N_t}$ is expressed as [53]

$$\mathbf{C}_t = \text{Toeplitz}\{\mathbf{c}_t, 0, \dots, 0\}, \quad (3)$$

where $\mathbf{c}_t = [1, c_{t,1}, \dots, c_{t,P-1}]^T \in \mathbb{C}^P$ is the MC vector and $c_{t,p}$ denotes the p -th non-zero element with $p = 1, 2, \dots, P-1$. Given the reduced coupling effects due to the increased distance between the element distance, we constrain $0 < |c_{t,P-1}| < \dots < |c_{t,1}| < |c_{t,0}| = 1$. Also, the MC matrix of the receiver antenna array $\mathbf{C}_r \in \mathbb{C}^{N_r \times N_r}$ can be obtained in a similar way based on the MC vector $\mathbf{c}_r \in \mathbb{C}^P$. The symmetric Toeplitz matrix is employed in our work for two reasons. First, it is physically justified, as its structure accurately reflects the coupling properties inherent to ULAs [54], [55]. Second, compared to the arbitrary matrix model with N_t^2 unknown parameters, it offers significant computational tractability by drastically reducing the number of model parameters, which is crucial for efficient feature estimation.

2) Statistical Model of MC Fingerprint: Following the works in [30], [53], the gain and phase terms of $c_{t,p} = c_{t,p}^G \exp(jc_{t,p}^I)$

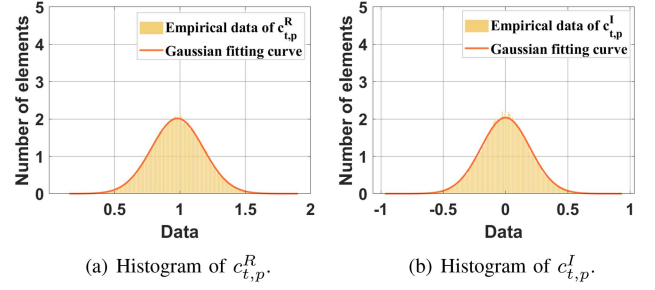


Fig. 2. Histogram of $c_{t,p}^R$ and $c_{t,p}^I$ with Gaussian fitting curves.

are both modeled by the Gaussian distribution, that is, $c_{t,p}^G \sim \mathcal{N}(\frac{1}{2d_p}, (\frac{1}{10d_p})^2)$ and $c_{t,p}^I \sim \mathcal{N}(0, (\frac{\pi}{18})^2)$. d_p denotes the normalized antenna spacing distance between the first and the p -th element. Accordingly, we implement 10^5 simulation trials to realize $c_{t,p}$ and to explore its statistical characteristics by observing the histogram of the results. Fig. 2(a) and (b) present the histogram results of the real and imaginary parts concerning $c_{t,p}$ (denoted by $c_{t,p}^R$ and $c_{t,p}^I$, respectively) with Gaussian theoretical curves. It is clear that the empirical data of both $c_{t,p}^R$ and $c_{t,p}^I$ are nicely fitted by the Gaussian curves, indicating that we can leverage the Gaussian distribution model to effectively depict the statistical behaviors of $c_{t,p}^R$ and $c_{t,p}^I$. In addition, we also note that the variances of $c_{t,p}^R$ and $c_{t,p}^I$ are highly close to each other and $c_{t,p}^R$ and $c_{t,p}^I$ are statistically independent. Based on the above observations, we resort to the complex Gaussian distribution with mean vector $\bar{\mathbf{c}}_t$ and covariance matrix $\mathbf{K}_c$ to model the MC coefficient vector $\mathbf{c}_t$, that is,

$$p(\mathbf{c}_t; \bar{\mathbf{c}}_t, \mathbf{K}_c) = \frac{1}{\pi^P |\mathbf{K}_c|} \exp(-(\mathbf{c}_t - \bar{\mathbf{c}}_t)^H \mathbf{Q}_c (\mathbf{c}_t - \bar{\mathbf{c}}_t)), \quad (4)$$

where $\bar{\mathbf{c}}_t = [\bar{c}_{t,0}, \bar{c}_{t,1}, \dots, \bar{c}_{t,P-1}]^T$ and $\mathbf{K}_c = \sigma_c^2 \mathbf{I}$. Moreover, $\mathbf{Q}_c = \mathbf{K}_c^{-1}$ denotes the inverse matrix of $\mathbf{K}_c$.

3) Feasibility Analysis of MC Fingerprint: Distinguishability and stability are important criteria for measuring the feasibility of the MC feature, which are discussed as follows.

Distinguishability of MC fingerprint: The MC effects in antenna arrays are determined by a combination effect of array design, element layout, material properties, and minor manufacturing tolerances [29], [30]. These factors lead to varying degrees of mutual influence between each antenna element, resulting in a unique coupling pattern [56], [57]. This pattern is highly distinguishable between different antenna arrays, that is, different antenna arrays inherently possess different MC features.

Stability of MC fingerprint: The stability of MC effects mainly results from two aspects. On the one hand, the relatively stable physical structure of the antenna array, determined during the production process, makes it less susceptible to mechanical deformation. This ensures that the MC effect can remain stable over a large time scale. On the other hand, the manufacturing materials of antenna arrays often have characteristics of high-temperature resistance, wear resistance, and deformation resistance, which help to maintain the structural integrity of the antenna array (avoiding shifts in the position or deformation of elements). These characteristics can also reduce changes in MC effects caused by temperature variations.

E. AoA-Based Spatial Fingerprint Model

In comparison with previously proposed channel features like CIR, it is demonstrated in [28] that the AoA effectively reveals the fine-grained location information around the transmitter node and is highly expected to serve as the spatial fingerprint to uniquely characterize each device identity due to its excellent stability and elevated forgery difficulty. Since mmWave signals commonly propagate through a few clusters with relatively small angular spread, the individually resolvable path components will change at a significantly slow rate. More specifically, AoAs θ_l determined by the large-scale scattering environment tend to vary very slowly over the large time scale even though the aggregate channel matrix $\mathbf{H}$ exhibits more significant fluctuations. In addition, the small-scale mobility exerts little impact on the AoA, which has been validated through the experimental investigations in [58]. This phenomenon indicates that AoA-based spatial fingerprints can possess high stability during the authentication process. Furthermore, attackers usually need to be co-located within a few centimeters of legitimate UE to mimic sensitive AoA spatial fingerprints, which is highly challenging [59]. Given the above two characteristics of AoA, we resort to such a spatial feature to assist in designing a hybrid PHY-layer authentication scheme.

To comprehensively explore the authentication performance of AoA, two statistical models associated with different mmWave frequency bands are introduced as follows. For 28 GHz and 73 GHz mmWave frequency bands, the AoAs of multipath are mainly characterized by the angular spread $\sigma_{\theta_l, \mathcal{G}}$ with the Gaussian distribution [32]. If we denote $\bar{\theta}_l$ the AoA mean of the l -th propagation path, then the generated AoAs θ_l are conformed to a Gaussian distribution with variance $\sigma_{\theta_l, \mathcal{G}}^2$:

$$p(\theta_l; \bar{\theta}_l, \sigma_{\theta_l, \mathcal{G}}^2) = \frac{1}{\sqrt{2\pi\sigma_{\theta_l, \mathcal{G}}^2}} \exp\left(-\frac{(\theta_l - \bar{\theta}_l)^2}{2\sigma_{\theta_l, \mathcal{G}}^2}\right). \quad (5)$$

Empirical observations indicate that AoA distributions in the 60 GHz band, particularly in indoor and industrial deployment scenarios [32], [33], often exhibit distinct non-Gaussian traits, notably sharper peaks around the mean and heavier tails compared to a Gaussian profile with equivalent variance. These characteristics are primarily attributed to 60 GHz propagation dynamics, where dominant line-of-sight (LoS) or strong specular multipath components contribute to pronounced peaks, and distinct scattering phenomena result in heavier tails. We note that the Laplace distribution, with its double exponential form, is mathematically well-suited to represent data with these specific traits. Furthermore, channel characterization studies targeting indoor WLAN environments [32] and complex warehouse settings [33] have empirically demonstrated that the Laplace distribution more accurately captures these observed AoA statistics, especially for intra-cluster angular parameters, than a traditional Gaussian model. Consequently, we employ the Laplace distribution, with its inherent ability to model such peaked and heavy-tailed data, to characterize the statistical model of the AoA feature. If we use $\sigma_{\theta_l, \mathcal{L}}$ to denote the scale parameter reflecting the impact of angular spread, then the probability density function (PDF) of a Laplace random variable θ_l with mean $\bar{\theta}_l$ is mathematically written as

$$p(\theta_l; \bar{\theta}_l, \sigma_{\theta_l, \mathcal{L}}) = \frac{1}{2\sigma_{\theta_l, \mathcal{L}}} \exp\left(-\frac{|\theta_l - \bar{\theta}_l|}{\sigma_{\theta_l, \mathcal{L}}}\right). \quad (6)$$

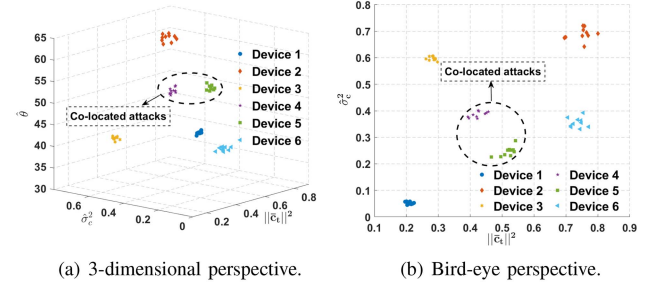


Fig. 3. Feature space visualization of the proposed two-modal fingerprints.

For convenience, we conduct the authentication performance analysis with the settings as $\sigma_{\theta_1, \mathcal{G}}^2 \approx \sigma_{\theta_2, \mathcal{G}}^2 \approx \dots \approx \sigma_{\theta_L, \mathcal{G}}^2 = \sigma_{\theta, \mathcal{G}}^2$ and $\sigma_{\theta_1, \mathcal{L}} \approx \sigma_{\theta_2, \mathcal{L}} \approx \dots \approx \sigma_{\theta_L, \mathcal{L}} = \sigma_{\theta, \mathcal{L}}$.

F. Feasibility Illustration of Two-Modal Features

To illustrate the feasibility of the employed MC and AoA fingerprints, we consider 6 communication devices (namely Device 1-6) and plot in Fig. 3 the feature extraction results about the estimated $\hat{\sigma}_c^2$, $\|\bar{\mathbf{c}}_t\|^2$, and $\hat{\theta}$ from 3-dimensional and bird-eye perspectives. As clearly observed from Fig. 3(a) and (b), different transmitters are governed by different physical parameters. Furthermore, one can also note that the physical layer features corresponding to the same device are settled within a certain range, revealing the stability of the proposed two-modal fingerprints. We also evaluate the feasibility of the proposed authentication scheme under the co-located attack scenario, where Device 4 and Device 5 are positioned near each other (within a few centimeters). Fig. 3(a) shows that the AoA features associated with Device 4 and Device 5 are very similar, implying that the AoA fingerprint may confront challenges in distinguishing devices when the attacker is highly close to the legal user. Fortunately, in Fig. 3(b), the proposed authentication scheme can still resort to the intrinsic MC hardware feature to uniquely characterize the device identities. These results imply that the adopted two-modal features are feasible and effective even when faced with a co-located attack.

G. Communication Model With Two-Modal Features

If we use $\mathbf{s}(n, k)$ and $\tilde{\mathbf{v}}(n, k)$ to denote the orthogonal transmitting signal and the noise at instant n of frame k ($n = 1, 2, \dots, N$), respectively, then the signal $\tilde{\mathbf{y}}(n, k)$ at the receiver side is written as

$$\tilde{\mathbf{y}}(n, k) = \sum_{l=1}^L \alpha_l(n, k) [\mathbf{C}_r \mathbf{a}_r(\theta_l)] [\mathbf{C}_t \mathbf{a}_t(\phi_l)]^H \mathbf{s}(n, k) + \tilde{\mathbf{v}}(n, k). \quad (7)$$

After conducting orthogonal matching of $\tilde{\mathbf{y}}(n, k)$ and stacking all the match-filtered signals into one column, we can obtain the final received signal $\mathbf{y}(n, k) \in \mathbb{C}^{N_r N_t \times 1}$, given by

$$\mathbf{y}(n, k) = [\tilde{\mathbf{y}}_1^T(n, k), \dots, \tilde{\mathbf{y}}_2^T(n, k), \dots, \tilde{\mathbf{y}}_{N_t}^T(n, k)]^T = \mathbf{C} \Phi(\theta, \phi) \boldsymbol{\alpha}(n, k) + \mathbf{v}(n, k), \quad (8)$$

where $\tilde{\mathbf{y}}_m^T(n, k) = \tilde{\mathbf{y}}(n, k) \mathbf{s}_m^H(n, k) \in \mathbb{C}^{N_r}$ represents the distinguished signal using the m -th matched filter $\mathbf{s}_m^H(n, k)$ and $\mathbf{C} = \mathbf{C}_r \otimes \mathbf{C}_t \in \mathbb{C}^{N_r N_t \times N_r N_t}$ is the MC matrix. $\mathbf{v}(n, k)$ is the complex Gaussian noise with variance σ_v^2 , that is, $\mathbf{v}(n, k) \sim$

$\mathcal{CN}(0, \sigma_v^2 \mathbf{I})$. $\alpha(n, k) = [\alpha_1(n, k), \alpha_2(n, k), \dots, \alpha_L(n, k)]^T \in \mathbb{C}^L$ denotes the complex gain vector and $\Phi(\theta, \phi) \in \mathbb{C}^{N_r N_t \times L}$ denotes the ideal steering matrix, further expressed by

$$\begin{aligned} \Phi(\theta, \phi) &= [\mathbf{a}_r(\theta_1), \dots, \mathbf{a}_r(\theta_L)] \odot [\mathbf{a}_t(\phi_1), \dots, \mathbf{a}_t(\phi_L)] \\ &= \underbrace{\mathbf{A}_r(\theta) \odot \mathbf{A}_t(\phi)}_{\mathbf{A}}. \end{aligned} \quad (9)$$

Based on (8) and (9), we rewrite the signal model involving MC and AoA spatial fingerprints in a more compact fashion, i.e.,

$$\mathbf{y}(n, k) = \mathbf{C}\mathbf{A}\alpha(n, k) + \mathbf{v}(n, k). \quad (10)$$

IV. SECURE TWO-MODAL AUTHENTICATION SCHEME

A. Feature Extraction

In this section, we employ the eigenspace-based multiple signal classification (MUSIC) algorithm [60] to jointly extract the MC and AoA features. We use $\hat{\mathbf{R}}$, $\mathbf{U}_s$, and $\mathbf{U}_v$ to denote the estimated covariance matrix, the signal subspace and the noise subspace, respectively. When N snapshots are available, $\hat{\mathbf{R}}$ can be obtained as

$$\begin{aligned} \hat{\mathbf{R}} &= \frac{1}{N} \sum_{n=1}^N \mathbf{y}(n, k) \mathbf{y}^H(n, k) \\ &= \mathbf{U}_s \mathbf{\Lambda}_s \mathbf{U}_s^H + \mathbf{U}_v \mathbf{\Lambda}_v \mathbf{U}_v^H, \end{aligned} \quad (11)$$

where $\mathbf{\Lambda}_s$ and $\mathbf{\Lambda}_v$ represent two diagonal matrices containing L dominate eigenvalues and the rest eigenvalues, respectively. Since $\mathbf{U}_s$ is orthogonal to $\mathbf{U}_v$ and $\mathbf{U}_s$ span the same subspace with the columns of $\mathbf{C}\mathbf{A}$, we can compute the MUSIC spectrum of the l -th spatial angle like

$$\begin{aligned} \mathbf{P}(\theta_l) &= ([\mathbf{C}_r \mathbf{a}_r(\theta_l) \otimes \mathbf{C}_t \mathbf{a}_t(\phi_l)]^H \mathbf{U}_v \\ &\quad \times \mathbf{U}_v^H [\mathbf{C}_r \mathbf{a}_r(\theta_l) \otimes \mathbf{C}_t \mathbf{a}_t(\phi_l)])^{-1}. \end{aligned} \quad (12)$$

Then, the AoA measurements are calculated by searching L largest peaks of (12). However, the adopted MUSIC algorithm is invalid when both $\mathbf{C}_r$ and $\mathbf{C}_t$ are unknown. To this end, we first introduce the following Lemma 1 to obtain such MC effects.

Lemma 1: For a symmetric Toeplitz matrix $\mathbf{Z} \in \mathbb{C}^{M \times M}$ and a vector $\mathbf{a} \in \mathbb{C}^M$, if $\mathbf{Z}$ contains P distinct elements $\mathbf{z} = [z_1, z_2, \dots, z_P]^T \in \mathbb{C}^P$ ($P < M$), then the following data transformation is established [53], [55]:

$$\mathbf{Z}\mathbf{a} = \mathbf{T}\mathbf{z}, \quad (13)$$

where the p -th column of $\mathbf{T}$ is given by $[\mathbf{T}]_{:,p} = \mathbf{J}_p \mathbf{a}$ and $\mathbf{J}_p$ is further written as

$$[\mathbf{J}_p]_{i,j} = \begin{cases} 1, & \text{if } [\mathbf{Z}]_{i,j} = z_p, \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

Hence, based on the Lemma 1, we can transform $\mathbf{C}_t \mathbf{a}_t(\phi_l)$ and $\mathbf{C}_r \mathbf{a}_r(\theta_l)$ into

$$\mathbf{C}_t \mathbf{a}_t(\phi_l) = \mathbf{T}_t(\phi_l) \mathbf{c}_t, \quad \mathbf{C}_r \mathbf{a}_r(\theta_l) = \mathbf{T}_r(\theta_l) \mathbf{c}_r. \quad (15)$$

Then, the AoA and MC fingerprints are extracted by the following optimization problem:

$$\min_{\mathbf{c}_{rt}} \underbrace{\mathbf{c}_{rt}^H [\mathbf{T}_r(\theta_l) \otimes \mathbf{T}_t(\phi_l)]^H \mathbf{U}_v \mathbf{U}_v^H [\mathbf{T}_r(\theta_l) \otimes \mathbf{T}_t(\phi_l)] \mathbf{c}_{rt}}_{\Psi(\theta_l, \phi_l)}, \quad (16)$$

where $\mathbf{c}_{rt} = \mathbf{c}_r \otimes \mathbf{c}_t \in \mathbb{C}^{P^2}$. It is clear that (16) is a quadratic optimization problem. To avoid the trivial solution that $\mathbf{c}_{rt} = \mathbf{0}$, we enforce a constraint $\mathbf{g}^H \mathbf{c}_{rt} = \xi$, where ξ is a non-zero

constant and $\mathbf{g} = [1, 0, \dots, 0]^T \in \mathbb{C}^{P^2}$. Then, the problem (16) is transformed to

$$\min_{\mathbf{c}_{rt}} \underbrace{\mathbf{c}_{rt}^H [\mathbf{T}_r(\theta_l) \otimes \mathbf{T}_t(\phi_l)]^H \mathbf{U}_v \mathbf{U}_v^H [\mathbf{T}_r(\theta_l) \otimes \mathbf{T}_t(\phi_l)] \mathbf{c}_{rt}}_{\Psi(\theta_l, \phi_l)}, \quad (17)$$

$$\text{s.t. } \mathbf{g}^H \mathbf{c}_{rt} = \xi. \quad (17a)$$

The above problem can be addressed by the Lagrange multiplier method. If we use η to denote the Lagrange multiplier, the Lagrange function is constructed as

$$\mathcal{F}(\theta_l, \phi_l) = \mathbf{c}_{rt}^H \Psi(\theta_l, \phi_l) \mathbf{c}_{rt} - \eta(\mathbf{g}^H \mathbf{c}_{rt} / \xi - 1). \quad (18)$$

By taking the partial derivative of (18) with respect to $\mathbf{c}_{rt}$ to zero, we can obtain

$$\mathbf{c}_{rt} = \frac{\xi \Psi^{-1}(\theta_l, \phi_l) \mathbf{g}}{\mathbf{g}^H \Psi^{-1}(\theta_l, \phi_l) \mathbf{g}}. \quad (19)$$

After inserting (19) into (16), the final optimization problem is expressed by

$$\hat{\theta}, \hat{\phi} = \arg \max_{\theta, \phi} \mathbf{g}^H \Psi^{-1}(\theta_l, \phi_l) \mathbf{g}. \quad (20)$$

Then, we conduct the grid search to find the L peaks of (20) for the AoA-AoD pair. Based on the estimated $\hat{\theta}$ and $\hat{\phi}$, the estimation of $\mathbf{c}_{rt}$ denoted by $\hat{\mathbf{c}}_{rt}$ is formulated as

$$\hat{\mathbf{c}}_{rt} = \frac{1}{L} \sum_{l=1}^L \frac{\xi \Psi^{-1}(\hat{\theta}_l, \hat{\phi}_l) \mathbf{g}}{\mathbf{g}^H \Psi^{-1}(\hat{\theta}_l, \hat{\phi}_l) \mathbf{g}}. \quad (21)$$

Finally, we can extract the MC feature $\hat{\mathbf{c}}_t$ related to the transmitter from $\hat{\mathbf{c}}_{rt}$ as

$$\hat{c}_{t,p} = \hat{c}_{rt,p}, \quad p = 1, 2, \dots, P. \quad (22)$$

Considering the inevitable estimation error resulting from the background noise, we decompose estimations of AoA and MC features as the sum of the true value and a Gaussian estimation error [28], shown respectively as

$$\hat{\theta}_l = \theta_l + w_\theta, \quad \hat{c}_{t,p} = c_{t,p} + w_c, \quad (23)$$

where w_θ is a zero-mean Gaussian variable with variance $\sigma_{w,\theta}^2$ and w_c presents a zero-mean complex Gaussian variable with variance $\sigma_{w,c}^2$. We determine the mean square error as the value of $\sigma_{w,\theta}^2$ and $\sigma_{w,c}^2$, and discuss the theoretical lower bounds of $\sigma_{w,\theta}^2$ and $\sigma_{w,c}^2$ in Section V-C.

B. Two-Modal Decision Criterion

In this subsection, we present the identity validation criteria tailored for both Gaussian/Laplace AoA scenarios.

1) *Identity Validation Framework Under Gaussian AoA:* Based on the extracted two-modal device fingerprints in terms of MC coefficients and AoA, Bob needs to determine the origin of the current received signal (from Alice or Eve). Let A and E denote Alice and Eve, respectively. Then, the device identity validation can be formulated as a binary hypothesis testing involving the null hypothesis $\mathcal{H}_0$ and the alternate hypothesis $\mathcal{H}_1$, detailed as below:

$$\begin{cases} \mathcal{H}_0 : \hat{\boldsymbol{\theta}} \sim \mathcal{N}(\bar{\boldsymbol{\theta}}_A, \tilde{\sigma}_{\theta_A, g}^2 \mathbf{I}), \hat{\mathbf{c}}_t \sim \mathcal{CN}(\bar{\mathbf{c}}_{t,A}, \tilde{\sigma}_{c,A}^2 \mathbf{I}), \\ \mathcal{H}_1 : \hat{\boldsymbol{\theta}} \sim \mathcal{N}(\bar{\boldsymbol{\theta}}_E, \tilde{\sigma}_{\theta_E, g}^2 \mathbf{I}), \hat{\mathbf{c}}_t \sim \mathcal{CN}(\bar{\mathbf{c}}_{t,E}, \tilde{\sigma}_{c,E}^2 \mathbf{I}), \end{cases} \quad (24)$$

where $\mathcal{H}_0$ represents that the current transmitter is Alice and $\mathcal{H}_1$ indicates that the received signal originates from Eve. Due to the Gaussian additivity, it is obvious that $\hat{\boldsymbol{\theta}}$ is a Gaussian vector with mean $\bar{\boldsymbol{\theta}}_A = [\bar{\theta}_{1,A}, \bar{\theta}_{2,A}, \dots, \bar{\theta}_{L,A}]^T$ and variance

$\tilde{\sigma}_{\theta_A, \mathcal{G}}^2 = \sigma_{\theta_A, \mathcal{G}}^2 + \sigma_{w, \theta}^2$ under hypothesis $\mathcal{H}_0$. Also, $\hat{\mathbf{c}}_t$ obeys a complex Gaussian distribution with mean $\bar{\mathbf{c}}_{t, A}$ and variance $\tilde{\sigma}_{c, A}^2 = \sigma_{c, A}^2 + \sigma_{w, c}^2$. Correspondingly, on $\mathcal{H}_1$, $\hat{\boldsymbol{\theta}}$ also follows a Gaussian distribution with mean $\bar{\boldsymbol{\theta}}_E = [\bar{\theta}_{1, E}, \bar{\theta}_{2, E}, \dots, \bar{\theta}_{L, E}]^T$ and variance $\tilde{\sigma}_{\theta_E, \mathcal{G}}^2 = \sigma_{\theta_E, \mathcal{G}}^2 + \sigma_{w, \theta}^2$ and $\hat{\mathbf{c}}_t$ is a complex Gaussian vector with mean $\bar{\mathbf{c}}_{t, E}$ and variance $\tilde{\sigma}_{c, E}^2 = \sigma_{c, E}^2 + \sigma_{w, c}^2$.

Design of the LWC Authentication Scheme: To implement the authentication decision in (24), we first utilize the likelihood ratio test (LRT) framework to establish two identity detectors with only AoA and MC feature, respectively. Then, we assign the feature weights to the developed detectors to design the final two-modal LWC authentication scheme.

Lemma 2: Given the PDF of Gaussian AoA in (5), the log-LRT detector with respect to $\hat{\boldsymbol{\theta}}$ is evaluated as [61]

$$\Lambda(\hat{\boldsymbol{\theta}}) = \frac{1}{2} \boldsymbol{\theta}_{\Delta}^T \boldsymbol{\theta}_{\Delta} \underset{H_0}{\overset{H_1}{\geq}} \tau_{\theta}, \quad (25)$$

where $\Lambda(\cdot)$ denotes the LRT operator and $\boldsymbol{\theta}_{\Delta} = (\hat{\boldsymbol{\theta}} - \mathbf{m}_{\theta}) / \tilde{\sigma}_{\theta_m}$ with $\mathbf{m}_{\theta} = \tilde{\sigma}_{\theta_m}^2 (\bar{\boldsymbol{\theta}}_A / \tilde{\sigma}_{\theta_A, \mathcal{G}}^2 - \bar{\boldsymbol{\theta}}_E / \tilde{\sigma}_{\theta_E, \mathcal{G}}^2)$ and $\tilde{\sigma}_{\theta_m}^2 = \tilde{\sigma}_{\theta_A, \mathcal{G}}^2 \tilde{\sigma}_{\theta_E, \mathcal{G}}^2 / (\tilde{\sigma}_{\theta_E, \mathcal{G}}^2 - \tilde{\sigma}_{\theta_A, \mathcal{G}}^2)$. In addition, τ_{θ} is a preset threshold of the AoA spatial fingerprint.

Proof: See Appendix A, available online for the derivation process. ■

Following similar steps, the LRT detector using the MC fingerprint is evaluated by

$$\Lambda(\hat{\mathbf{c}}_t) = \mathbf{c}_{\Delta}^H \mathbf{c}_{\Delta} \underset{H_0}{\overset{H_1}{\geq}} \tau_c, \quad (26)$$

where $\mathbf{c}_{\Delta} = (\hat{\mathbf{c}}_t - \mathbf{m}_c) / \tilde{\sigma}_{c_m}$ with $\mathbf{m}_c = \tilde{\sigma}_{c_m}^2 (\bar{\mathbf{c}}_{t, A} / \tilde{\sigma}_{c, A}^2 - \bar{\mathbf{c}}_{t, E} / \tilde{\sigma}_{c, E}^2)$ and $\tilde{\sigma}_{c_m}^2 = \tilde{\sigma}_{c, A}^2 \tilde{\sigma}_{c, E}^2 / (\tilde{\sigma}_{c, E}^2 - \tilde{\sigma}_{c, A}^2)$. In addition, τ_c is a preset threshold with respect to MC fingerprint.

Based on the two-modal device identity discriminators in (25) and (26), we incorporate the idea of a linear weighted combination to design an effective authentication approach. If we use ω_1 and ω_2 to denote weight values assigned to AoA and MC features, respectively, then the final detector of the LWC scheme with the threshold $\tau_{\mathcal{G}}$ is constructed as

$$\Lambda(\hat{\boldsymbol{\theta}}, \hat{\mathbf{c}}_t) = \omega_1 2 \tilde{\sigma}_{\theta_m}^2 \Lambda(\hat{\boldsymbol{\theta}}) + \omega_2 2 \tilde{\sigma}_{c_m}^2 \Lambda(\hat{\mathbf{c}}_t) \underset{H_0}{\overset{H_1}{\geq}} \tau_{\mathcal{G}}. \quad (27)$$

By selecting the appropriate weight values, this detector facilitates the authentication process to prioritize the most discriminative feature and thus truly achieves an effective fusion of the two features.

2) Identity Validation Under Laplace AoA: Considering the Laplace AoA statistical characteristics under the 60 GHz frequency band, we note from (23) that $\hat{\theta}_l$ is a sum of a symmetric Laplace variable and a zero-mean Gaussian variable, which follows a Normal-Laplace distribution parameterized by the Laplace scale parameters $(\bar{\theta}_l$ and $\sigma_{\theta, \mathcal{L}})$ and Gaussian variance $\sigma_{w, \theta}^2$ (i.e., $\hat{\theta}_l \sim \mathcal{NL}(\bar{\theta}_l, \sigma_{\theta, \mathcal{L}}, \sigma_{w, \theta}^2)$) [62]. Similarly, the device identity validation is also constructed as a binary hypothesis test

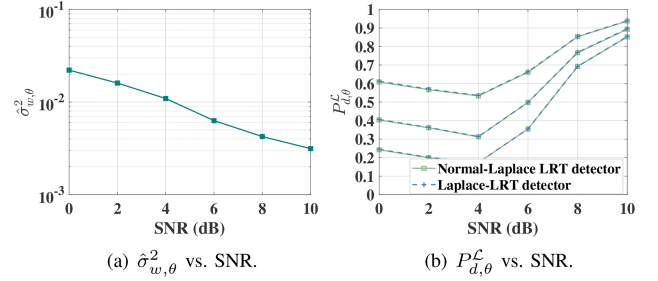


Fig. 4. Feasibility validation of the adopted Laplace approximation.

problem:

$$\begin{cases} \mathcal{H}_0 : \hat{\boldsymbol{\theta}} \sim \mathcal{NL}(\bar{\boldsymbol{\theta}}_A, \sigma_{\theta_A, \mathcal{L}} \mathbf{I}, \sigma_{w, \theta}^2 \mathbf{I}), \hat{\mathbf{c}}_t \sim \mathcal{CN}(\bar{\mathbf{c}}_{t, A}, \tilde{\sigma}_{c, A}^2 \mathbf{I}), \\ \mathcal{H}_1 : \hat{\boldsymbol{\theta}} \sim \mathcal{NL}(\bar{\boldsymbol{\theta}}_E, \sigma_{\theta_E, \mathcal{L}} \mathbf{I}, \sigma_{w, \theta}^2 \mathbf{I}), \hat{\mathbf{c}}_t \sim \mathcal{CN}(\bar{\mathbf{c}}_{t, E}, \tilde{\sigma}_{c, E}^2 \mathbf{I}). \end{cases} \quad (28)$$

If we use $\Lambda(\hat{\boldsymbol{\theta}}_{\mathcal{L}})$ and $\tau_{\theta, \mathcal{L}}$ to denote the LRT detector using the Normal-Laplace AoA feature and corresponding decision threshold, respectively, then $\Lambda(\hat{\boldsymbol{\theta}}_{\mathcal{L}})$ for the hypothesis testing in (28) is given in (29), shown at the bottom of this page.

It is observed that the detector in (29) exhibits relatively high computational complexity and potentially leads to unsatisfied time delay in mmWave MIMO systems. To this end, the Laplace approximation is utilized to simplify the complex Normal-Laplace detector into a Laplace model-based one. In other words, we approximate $\hat{\theta}_l$ as a Laplace variable with a very small estimation error (close to 0) and then the LRT detector can be simplified as follows. If we use $\tau_{\theta, \mathcal{L}}$ to denote the decision threshold, the simplified LRT detector is given by

$$\Lambda(\hat{\boldsymbol{\theta}}_{\mathcal{L}}) = \sum_{l=1}^L \left(\frac{|\hat{\theta}_l - \bar{\theta}_{A, l}|}{\sigma_{\theta_A, \mathcal{L}}} - \frac{|\hat{\theta}_l - \bar{\theta}_{E, l}|}{\sigma_{\theta_E, \mathcal{L}}} \right) \underset{H_0}{\overset{H_1}{\geq}} \tau_{\theta, \mathcal{L}}. \quad (30)$$

To validate the feasibility of the adopted Laplace approximation method, Fig. 4(a) and (b) illustrate how the estimated variance of the AoA estimation error (denoted by $\hat{\sigma}_{w, \theta}^2$) and detection performance (denoted by $P_{d, \theta}^{\mathcal{L}}$) vary with SNR, respectively. From Fig. 4(a), it can be observed that $\hat{\sigma}_{w, \theta}^2$ presents a declining tendency and is gradually close to 0 as SNR increases. Meanwhile, one can also note in Fig. 4(b) that the detection performance of the simplified LRT detector based on the Laplace distribution is aligned with that using the Normal-Laplace distribution conditioned on three different decision thresholds. This indicates that the performance degradation of the simplified detector is exceedingly marginal and demonstrates the feasibility and practicality of the Laplace approximation. Consequently, we leverage the Laplace-based LRT detector in (30) to devise the authentication framework with lower computational complexity and time overhead. Herein, we introduce a new TTC device authentication method under the Laplace AoA condition.

Design of the TTC Authentication Scheme: For the TTC scheme, Bob first needs to separately utilize the developed two detectors for identity judgment and then collaboratively validate

$$\Lambda(\hat{\boldsymbol{\theta}}_{\mathcal{L}}) = \sum_{l=1}^L \ln \left[\frac{\exp \left(\frac{\sigma_{w, \theta}^2}{2 \sigma_{\theta_E, \mathcal{L}}^2} - \frac{\hat{\theta}_l - \bar{\theta}_{E, l}}{\sigma_{\theta_E, \mathcal{L}}} \right) Q \left(\frac{\sigma_{w, \theta}}{\sigma_{\theta_E, \mathcal{L}}} - \frac{\hat{\theta}_l - \bar{\theta}_{E, l}}{\sigma_{w, \theta}} \right) - \exp \left(\frac{\sigma_{w, \theta}^2}{2 \sigma_{\theta_A, \mathcal{L}}^2} + \frac{\hat{\theta}_l - \bar{\theta}_{E, l}}{\sigma_{\theta_E, \mathcal{L}}} \right) Q \left(-\frac{\sigma_{w, \theta}}{\sigma_{\theta_E, \mathcal{L}}} - \frac{\hat{\theta}_l - \bar{\theta}_{E, l}}{\sigma_{w, \theta}} \right)}{\exp \left(\frac{\sigma_{w, \theta}^2}{2 \sigma_{\theta_A, \mathcal{L}}^2} - \frac{\hat{\theta}_l - \bar{\theta}_{A, l}}{\sigma_{\theta_A, \mathcal{L}}} \right) Q \left(\frac{\sigma_{w, \theta}}{\sigma_{\theta_A, \mathcal{L}}} - \frac{\hat{\theta}_l - \bar{\theta}_{A, l}}{\sigma_{w, \theta}} \right) - \exp \left(\frac{\sigma_{w, \theta}^2}{2 \sigma_{\theta_A, \mathcal{L}}^2} + \frac{\hat{\theta}_l - \bar{\theta}_{A, l}}{\sigma_{\theta_A, \mathcal{L}}} \right) Q \left(-\frac{\sigma_{w, \theta}}{\sigma_{\theta_A, \mathcal{L}}} - \frac{\hat{\theta}_l - \bar{\theta}_{A, l}}{\sigma_{w, \theta}} \right)} \right] \underset{H_0}{\overset{H_1}{\geq}} \tau_{\mathcal{NL}}. \quad (29)$$

Algorithm 1: Synergistic Authentication Framework Under Gaussian and Laplace Scenarios.

Input: Received signal $\mathbf{y}(n, k)$, number of snapshots N , number of paths L , thresholds τ_θ , $\tau_{\theta,L}$, τ_c , τ_G , false alarm constraint Γ .

Output: Authentication decision $\mathcal{H}_0$ (Legitimate) or $\mathcal{H}_1$ (Attacker).

- 1: **Step 1: Two-Modal Feature Extraction**
 - 2: Estimate covariance matrix $\hat{\mathbf{R}}$ from $\mathbf{y}(n, k)$ using (11).
 - 3: Obtain $\mathbf{U}_s$ and $\mathbf{U}_v$ via eigendecomposition of $\hat{\mathbf{R}}$.
 - 4: Estimate AoA-AoD pair $(\hat{\theta}_l, \hat{\phi}_l)$, $l = 1, \dots, L$, by grid search using (20).
 - 5: Compute $\hat{\mathbf{c}}_{rt}$ by (21) and extract $\hat{\mathbf{c}}_t$ by (22).
 - 6: **Step 2: Identity Decision**
 - 7: **if** Gaussian AoA scenario **then**
 - 8: Compute the individual LRT statistics $\Lambda(\hat{\boldsymbol{\theta}})$ and $\Lambda(\hat{\mathbf{c}}_t)$ for AoA and MC features using (25) and (26).
 - 9: Optimal Weight Allocation: Solve problem in (54) using the SQP algorithm under $P_f^w = \Gamma$ to obtain the optimal feature weights ω_1 and ω_2 .
 - 10: Construct $\Lambda(\hat{\boldsymbol{\theta}}, \hat{\mathbf{c}}_t)$ with optimal weights using (27).
 - 11: Accept $\mathcal{H}_0$ if $\Lambda(\hat{\boldsymbol{\theta}}, \hat{\mathbf{c}}_t) < \tau_G$; otherwise, reject and declare $\mathcal{H}_1$.
 - 12: **else if** Laplace AoA scenario **then**
 - 13: Compute $\Lambda(\hat{\mathbf{c}}_t)$ and $\Lambda(\hat{\boldsymbol{\theta}}_L)$ using (26) and (30), respectively.
 - 14: Accept $\mathcal{H}_0$ only if $\Lambda(\hat{\boldsymbol{\theta}}_L) < \tau_{\theta,L}$ and $\Lambda(\hat{\mathbf{c}}_t) < \tau_c$; otherwise, declare $\mathcal{H}_1$.
 - 15: **end if**
 - 16: **return** final authentication decision
-

the current source of the received signal based on the outcomes of these two detectors. In specific, only when the decision results of both detectors are determined as $\mathcal{H}_0$, Bob will consider the current transmitter as the legitimate one. Otherwise, the received signals are deemed as originating from an unauthorized device and will be discarded. The TTC authentication decision can be further mathematically formulated as

$$\begin{cases} \mathcal{H}_0 : \Lambda(\hat{\boldsymbol{\theta}}_L) < \tau_{\theta,L} \text{ and } \Lambda(\hat{\mathbf{c}}_t) < \tau_c, \\ \mathcal{H}_1 : \text{Otherwise.} \end{cases} \quad (31)$$

Based on the developed feature extraction method and the corresponding decision criteria under Gaussian and Laplace AoA scenarios, the overall authentication procedure is summarized in Algorithm 1.

C. Computational Cost Analysis

Both LWC and TTC scheme comprises feature extraction and identity decision phase. The computational complexity of the feature extraction phase is determined by the following steps: covariance matrix estimation in (11), eigendecomposition, spectrum searching for AoA-AoD pair via (20), and MC feature estimation in (21). In specific, the computational complexity of the covariance matrix estimation from N signal snapshots is $\mathcal{O}(NM^2)$, where $M = N_r N_t$ is the total antenna dimension. The subsequent eigendecomposition of $\hat{\mathbf{R}}$ has a complexity of $\mathcal{O}(M^3)$. Let $N_{\text{grid}} = \frac{\Phi}{\delta}$ denote the total number of angular grid points searched for AoA-AoD pair estimation, where Φ

is the search range and δ is the grid size. The AoA feature extraction process identifies L paths by maximizing the objective function in (20) through a search over N_{grid} angular grid points (θ, ϕ) . The complexity for the peak search of each grid point is $\mathcal{O}(M^2 P^2 + MP^4 + P^6)$ and the total complexity of spectrum searching is $N_{\text{grid}} \mathcal{O}(M^2 P^2 + MP^4 + P^6)$. This indicates that the computational burden is inversely proportional to the grid size δ . After identifying the optimal AoA-AoD pair, the MC feature is estimated using (21) with a complexity of $\mathcal{O}(LP^4)$. Consequently, the total computational complexity for the two-modal feature estimation is $\mathcal{O}(NM^2 + M^3 + N_{\text{grid}}(M^2 P^2 + MP^4 + P^6) + LP^4)$. For the subsequent identity decision phase, the complexity varies by different schemes. The TTC scheme relies on simple threshold comparisons with a minimal complexity of $\mathcal{O}(L + P)$. The LWC scheme incorporates an additional sequential quadratic programming (SQP)-based weight optimization. In each SQP iteration, a quadratic programming subproblem with n variables is solved, typically incurring a complexity of $\mathcal{O}(n^3)$. Since our LWC scheme optimizes only $n = 2$ weights under a single linear constraint. This results in a total decision complexity of $\mathcal{O}(L + P + I_{\text{SQP}} \cdot n^3)$, where I_{SQP} is the number of SQP iterations. Moreover, the overall time cost of our schemes is approximately 10 ms (as detailed in Section VI-F), satisfying the low-latency authentication requirements in the concerned mmWave systems.

The proposed framework also demonstrates promising potential in large-scale antenna arrays and multi-user environments. Specifically, for arrays with $N_t \in \{64, 128\}$ elements, the increased computational complexity is effectively mitigated by narrowing the spatial search range Φ based on prior directional estimates from the initial beam alignment stage. This targeted searching drastically reduces N_{grid} , thereby offsetting the overhead of high-dimensional matrix operations and maintaining an acceptable total delay. Furthermore, since the AoA and MC fingerprints are link-specific and extracted from independent orthogonal pilot sequences, the authentication process remains fundamentally decoupled across multiple users. This facilitates parallel execution on modern hardware architectures, such as multi-core CPUs, GPUs, or FPGAs, to support concurrent secure link validation in high-density mmWave networks.

D. Security Performance Analysis

The robustness of the proposed scheme fundamentally relies on the unforgeable nature of physical hardware fingerprints. Although an adaptive adversary can forge the spatial AoA by dynamically steering beams, perfectly replicating the intrinsic MC fingerprint is challenging by two primary physical constraints [26]. First, even if Eve deploys an identical device model, she cannot circumvent the passive embedding of her own intrinsic MC characteristics into the transmitted signal. Digital phase manipulation aimed at spatial AoA spoofing cannot override or erase this physical residue, ensuring that Eve's unique hardware signature is invariably superimposed on the spoofed signals. Second, the MC effect constitutes a passive electromagnetic characteristic governed by microscopic, random manufacturing tolerances (e.g., element displacement and material impurities) specific to each physical unit [56]. Impersonating Alice's exact MC pattern would require Eve to exert precise microscopic manipulation over her own array's structural and electromagnetic properties, which is practically infeasible. Thus, based on the above observations, the proposed

scheme can effectively combat identity impersonation attacks, even against advanced adaptive strategies.

Remark 1: In practical antenna arrays, residual errors such as gain-phase mismatches and element-position uncertainties are often inevitable. Following the array error model in [63], [64], the received signal can be formulated as $\mathbf{y} = \mathbf{C}\tilde{\Gamma}\Delta(\theta)\mathbf{A}\alpha + \mathbf{v}$, where $\tilde{\Gamma}$ and $\Delta(\theta)$ represent the residual gain-phase error and position-induced perturbations, respectively. While residual calibration errors introduce an estimation error on the true MC coefficients, the extraction algorithm obtains an equivalent MC feature $\hat{\mathbf{C}}$ that encapsulates both the true MC and the stable residual calibration errors. Since these errors are uniquely tied to manufacturing tolerances, they serve as additional discriminative fingerprints rather than mere interference. This ensures the authentication scheme remains effective by leveraging a more comprehensive device-specific feature. Moreover, we further evaluate the sensitivity of proposed scheme to these non-ideal effects in Section VI-C.

V. THEORETICAL PERFORMANCE ANALYSIS

A. Modeling of the False Alarm and Detection Probabilities

1) *Theoretical Modeling Under Gaussian AoA Scenario:* A false alarm event denotes that the detector mistakenly identifies the legitimate device as an illegal one. In other words, under $\mathcal{H}_0$, the LRT detector decides the outcome as $\mathcal{H}_1$. Hence, the expression of false alarm only using the Gaussian AoA feature (denoted by $P_{f,\theta}^G$) is given in the Lemma 3.

Lemma 3: Given a threshold τ_θ , the analytical expression of $P_{f,\theta}^G$ is evaluated as

$$P_{f,\theta}^G = Q_{\chi_L^2(\lambda_0^\theta)} \left(\frac{2\tilde{\sigma}_{\theta_m}^2}{\tilde{\sigma}_{\theta_{A,G}}^2} \tau_\theta \right), \quad (32)$$

where $Q_{\chi_L^2(\lambda_0^\theta)}(\cdot)$ is the right-tail probability function for a Non-central Chi-square variable with degrees of freedom (DoF) L and noncentrality parameter λ_0^θ regarding the AoA fingerprint under $\mathcal{H}_0$. For a given τ , $Q_{\chi_L^2(\lambda_0^\theta)}(\tau)$ is further expressed by

$$Q_{\chi_L^2(\lambda_0^\theta)}(\tau) = \int_{\tau}^{\infty} \frac{1}{2} \left(\frac{x}{\lambda_0^\theta} \right)^{\frac{(L-2)}{4}} \exp \left(-\frac{1}{2} (x + \lambda_0^\theta) \right) I_{\nu} \left(\sqrt{x\lambda_0^\theta} \right) dx, \quad (33)$$

where $I_{\nu}(\cdot)$ denotes a modified Bessel function of the first kind with order $\nu = (L/2) - 1$.

Proof: See Appendix B, available online for the derivation process. ■

A detection event represents that Bob successfully determines that the received signal is from Alice, that is, the detector decides the outcome as $\mathcal{H}_1$ conditioned on $\mathcal{H}_1$. Then, the expression of detection probability only using the Gaussian AoA feature (denoted by $P_{d,\theta}^G$) is given as follows:

Given a threshold τ_θ , the analytical expression of $P_{d,\theta}^G$ is evaluated as

$$P_{d,\theta}^G = Q_{\chi_L^2(\lambda_1^\theta)} \left(\frac{2\tilde{\sigma}_{\theta_m}^2}{\tilde{\sigma}_{\theta_{E,G}}^2} \tau_\theta \right), \quad (34)$$

where $\lambda_1^\theta = (\mathbf{m}_1^\theta)^T \mathbf{m}_1^\theta$ with $\mathbf{m}_1^\theta = (\bar{\theta}_E - \mathbf{m}_\theta)/\tilde{\sigma}_{\theta_{E,G}}$ denotes the noncentrality parameter regarding the AoA fingerprint under $\mathcal{H}_1$.

For the LRT detector regarding the MC fingerprint, the corresponding false alarm and detection probabilities (denoted by P_f^c and P_d^c , respectively) are introduced as follows.

Lemma 4: For a given threshold τ_c , the expression of P_f^c is evaluated as

$$P_f^c = Q_{\chi_{2P}^2(\lambda_0^c)} \left(\frac{2\tilde{\sigma}_{c_m}^2}{\tilde{\sigma}_{c,A}^2} \tau_c \right), \quad (35)$$

where λ_0^c denotes the noncentrality parameter regarding the MC fingerprint under $\mathcal{H}_0$.

Proof: Similar to the process in Lemma 3, we make $\frac{\sqrt{2}\tilde{\sigma}_{c_m}}{\tilde{\sigma}_{c,A}} \mathbf{c}_\Delta | \mathcal{H}_0 \sim \mathcal{CN}(\frac{\sqrt{2}(\bar{\mathbf{c}}_{t,A} - \mathbf{m}_c)}{\tilde{\sigma}_{c,A}}, 2\mathbf{I})$ and use $\mathbf{m}_0^c = \frac{\sqrt{2}(\bar{\mathbf{c}}_{t,A} - \mathbf{m}_c)}{\tilde{\sigma}_{c,A}}$ to denote the mean of $\frac{\sqrt{2}\tilde{\sigma}_{c_m}}{\tilde{\sigma}_{c,A}} \mathbf{c}_\Delta | \mathcal{H}_0$. Let $\Lambda_0(\hat{\mathbf{c}}_t) = \frac{2\tilde{\sigma}_{c_m}^2}{\tilde{\sigma}_{c,A}^2} \mathbf{c}_\Delta^H \mathbf{c}_\Delta$ and we note that $\Lambda_0(\hat{\mathbf{c}}_t)$ is a Non-central Chi-square variable with noncentrality parameter $\lambda_0^c = (\mathbf{m}_0^c)^T \mathbf{m}_0^c$ and DoF $2P$ under $\mathcal{H}_0$. Based on the right-tail probability function of the Non-central Chi-square variable in (33), we can finally obtain the result in (35). ■

Given a threshold τ_c , the expression of P_d^c is evaluated as

$$P_d^c = Q_{\chi_{2P}^2(\lambda_1^c)} \left(\frac{2\tilde{\sigma}_{c_m}^2}{\tilde{\sigma}_{c,E}^2} \tau_c \right), \quad (36)$$

where $\lambda_1^c = (\mathbf{m}_1^c)^T \mathbf{m}_1^c$ denotes the noncentrality parameter regarding the MC fingerprint under $\mathcal{H}_1$ with $\mathbf{m}_1^c = \frac{\sqrt{2}(\bar{\mathbf{c}}_{t,E} - \mathbf{m}_c)}{\tilde{\sigma}_{c,E}}$.

Theoretical models of the TTC authentication scheme: First, we derive the theoretical models of the TTC scheme under the Gaussian scenario. On $\mathcal{H}_0$, the false alarm event occurs when at least one of the two individual detectors outputs $\mathcal{H}_1$. On $\mathcal{H}_1$, when $\Lambda(\hat{\theta}) > \tau_\theta$ or $\Lambda(\hat{\mathbf{c}}_t) > \tau_c$ holds, the detection event occurs. Accordingly, we derive the corresponding false alarm and detection probabilities (denoted by P_f^G and P_d^G , respectively) of the proposed TTC scheme in the following Theorem 1.

Theorem 1: For the considered mmWave MIMO communication system in Fig. 1, P_f^G and P_d^G of the TTC scheme under the Gaussian AoA scenario are evaluated as, respectively

$$P_f^G = P_{f,\theta}^G P_f^c + (1 - P_{f,\theta}^G) P_f^c + P_{f,\theta}^G (1 - P_f^c), \quad (37)$$

$$P_d^G = 1 - (1 - P_{d,\theta}^G)(1 - P_d^c). \quad (38)$$

Proof: Based on the equations in (32), (34), (35), and (36), we can easily obtain the above results. ■

Theoretical models of the LWC authentication scheme: Based on the LWC decision criterion in (27), the false alarm and detection probabilities (denoted by P_f^ω and P_d^ω , respectively) are defined as $\Pr(\Lambda(\hat{\theta}, \hat{\mathbf{c}}_t) > \tau_g | \mathcal{H}_0)$ and $\Pr(\Lambda(\hat{\theta}, \hat{\mathbf{c}}_t) > \tau_g | \mathcal{H}_1)$. To derive the expressions of P_f^ω and P_d^ω , it is imperative to investigate the PDFs of $\Lambda(\hat{\theta}, \hat{\mathbf{c}}_t)$ under two hypotheses. To this end, we provide the following Lemma 5 to acquire the PDFs of $\Lambda(\hat{\theta}, \hat{\mathbf{c}}_t) | \mathcal{H}_j$, $j = \{0, 1\}$.

Lemma 5: If the test statistic $T(x)$ is expressed as a weighted sum of W Noncentral Chi-square variables:

$$T(x) = \sum_{i=1}^W \omega_i \chi_{\nu_i}^2(\lambda_i), \quad (39)$$

where $\chi_{\nu_i}^2(\lambda_i)$ denotes the i -th Noncentral Chi-square variable with noncentrality parameter λ_i and DoF ν_i . Then, for a given threshold τ_x , the right-tail probability of $T(x)$ can be approximated by $Q_{\chi_{\nu_x}^2(\lambda_x)}(\tau_x^*)$ under a new threshold τ_x^* :

$$\Pr(T(x) > \tau_x) \approx Q_{\chi_{\nu_x}^2(\lambda_x)}(\tau_x^*), \quad (40)$$

where $\tau_x^* = [(\tau_x - \mu_T)/\sigma_T] \sqrt{2\nu_x + 4\lambda_x} + \nu_x + \lambda_x$. $\mu_T = \beta_1$ and $\sigma_T = \sqrt{2\beta_2}$ represent the mean and standard deviation of T with $\beta_j = \sum_{i=1}^W (\omega_i)^j \nu_i + j \sum_{i=1}^W (\omega_i)^j \lambda_i$. Moreover, λ_x and ν_x are respectively calculated by

$$\lambda_x = s_1 a^3 - a^2, \quad \nu_x = a^2 - 2\lambda_x, \quad (41)$$

where $a = 1/(s_1 - \sqrt{(s_1)^2 - s_2})$ with $s_1 = \beta_3/(\beta_2)^{3/2}$ and $s_2 = \beta_4/(\beta_2)^2$.

Proof: Please refer to [65] for the detailed proof. ■

Based on Lemma 5, we provide the analytical expressions of P_f^ω and P_d^ω in the following Theorem 2.

Theorem 2 Given a decision threshold τ_G , P_f^ω and P_d^ω of the LWC scheme are evaluated as, respectively

$$P_f^\omega \approx Q_{\chi_{\nu_0}^2(\lambda_0)}(\tau_{G,0}), \quad (42)$$

$$P_d^\omega \approx Q_{\chi_{\nu_1}^2(\lambda_1)}(\tau_{G,1}), \quad (43)$$

where $\tau_{G,i} = [(\tau_i - \mu_i^\omega)/\sigma_i^\omega] \sqrt{2\nu_i^\omega + 4\lambda_i^\omega} + \nu_i^\omega + \lambda_i^\omega$, $i = \{0, 1\}$ with $\tau_0 = \frac{\tau_G}{\sigma_{c,A}^2 \sigma_{\theta_A}^2}$ and $\tau_1 = \frac{\tau_G}{\sigma_{c,E}^2 \sigma_{\theta_E}^2}$.

Proof: See Appendix C, available online for the derivation process. ■

2) *Theoretical Modeling Under Laplace AoA Scenario:* To facilitate the establishment of a theoretical framework for the proposed TTC scheme under the Laplace AoA, we consider the scenario that Alice is positioned at the broadside of Bob's antenna array. This deployment ensures that the extracted AoA features follow a symmetric zero-mean Laplace distribution, i.e., $\theta_A = \theta_E = 0$. If we make $\tau_{\theta,\mathcal{L}}^* = \tau_{\theta,\mathcal{L}}(\frac{\sigma_{\theta_E,\mathcal{L}} - \sigma_{\theta_A,\mathcal{L}}}{\sigma_{\theta_A,\mathcal{L}} \sigma_{\theta_E,\mathcal{L}}})$, the LRT detector in (30) is written as

$$\Lambda(\hat{\theta}_{\mathcal{L}}) = \sum_{l=1}^L |\hat{\theta}_l| \stackrel{H_1}{\underset{H_0}{\gtrless}} \tau_{\theta,\mathcal{L}}^*. \quad (44)$$

Then, the expression of false alarm probability using the Laplace AoA feature (denoted by $P_{f,\theta}^{\mathcal{L}}$) is given in Lemma 6.

Lemma 6: Given a threshold $\tau_{\theta,\mathcal{L}}^*$, the analytical expression of $P_{f,\theta}^{\mathcal{L}}$ is evaluated as

$$P_{f,\theta}^{\mathcal{L}} = Q_{\chi_{2L}^2} \left(\frac{2}{\sigma_{\theta_A,\mathcal{L}}^2} \tau_{\theta,\mathcal{L}}^* \right), \quad (45)$$

where $Q_{\chi_{2L}^2}(\cdot)$ stands for the right tail probability function for a Central Chi-square variable with DoF $2L$ [61].

Proof: Due to the minor estimation error, $\hat{\theta}_l$ is approximated as a zero-mean symmetric Laplace random variable. Thus, on $\mathcal{H}_0$, $\frac{2}{\sigma_{\theta_A,\mathcal{L}}^2} \sum_{l=1}^L |\hat{\theta}_l|$ follows a Central Chi-square distribution with DoF $2L$. Consequently, $P_{f,\theta}^{\mathcal{L}}$ is derived by

$$\begin{aligned} P_{f,\theta}^{\mathcal{L}} &\triangleq \Pr \left(\sum_{l=1}^L |\hat{\theta}_l| > \tau_{\theta,\mathcal{L}}^* \middle| \mathcal{H}_0 \right) \\ &= \Pr \left(\frac{2}{\sigma_{\theta_A,\mathcal{L}}^2} \sum_{l=1}^L |\hat{\theta}_l| > \frac{2}{\sigma_{\theta_A,\mathcal{L}}^2} \tau_{\theta,\mathcal{L}}^* \right). \end{aligned} \quad (46)$$

We then substitute the $Q_{\chi_{2L}^2}(\cdot)$ function into (46) and finally obtain the result in (45). ■

Then, under a threshold $\tau_{\theta,\mathcal{L}}^*$, the expression of detection probability only using the Laplace AoA feature (denoted by $P_{d,\theta}^{\mathcal{L}}$) is given by

$$P_{d,\theta}^{\mathcal{L}} = Q_{\chi_{2L}^2} \left(\frac{2}{\sigma_{\theta_E,\mathcal{L}}^2} \tau_{\theta,\mathcal{L}}^* \right). \quad (47)$$

The final false alarm and detection probabilities for the TTC scheme under the Laplace AoA scenario are illustrated in Theorem 3.

Theorem 3: For the considered mmWave MIMO communication system in Fig. 1, $P_f^{\mathcal{L}}$ and $P_d^{\mathcal{L}}$ of the TTC scheme under Laplace AoA scenario are evaluated as, respectively

$$P_f^{\mathcal{L}} = P_{f,\theta}^{\mathcal{L}} P_f^c + (1 - P_{f,\theta}^{\mathcal{L}}) P_f^c + P_{f,\theta}^{\mathcal{L}} (1 - P_f^c), \quad (48)$$

$$P_d^{\mathcal{L}} = 1 - (1 - P_{d,\theta}^{\mathcal{L}})(1 - P_d^c). \quad (49)$$

Proof: Based on the equations in (35), (36), (45) and (47), we can easily obtain the above results. ■

To facilitate the practical deployment of the TTC scheme, the selection of the dual thresholds ($\tau_{\theta,\mathcal{L}}$ and τ_c) must be theoretically deterministic rather than purely empirical. Under the Neyman-Pearson criterion [61], the system specifies the tolerable false alarm probabilities for the spatial and hardware modalities, denoted as Γ_θ and Γ_c , respectively. By leveraging the theoretical false alarm models derived in Lemma 6 and Lemma 4, the practical thresholds can be calculated directly via the inverse complementary cumulative distribution functions:

$$\tau_{\theta,\mathcal{L}}^* = \frac{\sigma_{\theta_A,\mathcal{L}}}{2} Q_{\chi_{2L}^2}^{-1}(\Gamma_\theta), \quad \tau_c^* = \frac{\tilde{\sigma}_{c,A}}{2\tilde{\sigma}_{c,m}^2} Q_{\chi_{2P}^2(\lambda_0)}^{-1}(\Gamma_c). \quad (50)$$

This closed-form strategy provides the mmWave MIMO system with a flexible method to configure authentication strictness according to diverse security demands. Unfortunately, when the symmetric zero-mean assumption is relaxed, the test statistic in (30) involves the sum of L independent variables, each being the difference between two non-central Laplace terms. From a theoretical perspective, the resulting PDF consists of complex piecewise expressions involving transcendental functions, which prevents closed-form expressions for $P_{f,\theta}^{\mathcal{L}}$ and $P_{d,\theta}^{\mathcal{L}}$. Due to this analytical complexity, we have to rely on numerical simulations to evaluate the reliability of TTC scheme under this condition.

B. Optimal Weight Allocation Design

To ensure the optimal authentication performance of the proposed LWC scheme under Gaussian AoA, it is significantly crucial to choose the values of ω_1 and ω_2 . Accordingly, we first establish a function mapping relation between the weight value and the authentication performance metrics. Then, we carefully formulate the weight value determination as an optimization problem, acquiring the optimal feature weight that maximizes the detection probability under a fixed false alarm probability constraint. The detailed determination process is introduced as follows. Based on the Neyman-Pearson criterion [61], we first constrain a maximum tolerant false alarm probability $P_f^\omega = \Gamma$ and calculate the corresponding decision threshold $\tau_G^\dagger$:

$$\tau_{G,0}^\dagger = Q_{\chi_{\nu_0}^2(\lambda_0)}^{-1}(\Gamma), \quad (51)$$

$$\tau_G^\dagger = \tilde{\sigma}_{c,A}^2 \tilde{\sigma}_{\theta_A,G}^2 \left(\frac{\tau_{G,0}^\dagger - (\nu_0^\omega + \lambda_0^\omega)}{\sqrt{2\nu_0^\omega + 4\lambda_0^\omega}} \sigma_0^\omega + \mu_0^\omega \right). \quad (52)$$

Given the specified threshold $\tau_G^\dagger$, $P_d^{\omega,\dagger}$ is determined by

$$P_d^{\omega,\dagger} = Q_{\chi_{\nu_1}^2(\lambda_1^\omega)}(\tau_{G,1}^\dagger), \quad (53)$$

where $\tau_{G,1}^\dagger = [(\tau_1^\dagger - \mu_1^\omega)/\sigma_1^\omega] \sqrt{2\nu_1^\omega + 4\lambda_1^\omega} + \nu_1^\omega + \lambda_1^\omega$ and $\tau_1^\dagger = \frac{\tau_G^\dagger}{\tilde{\sigma}_{c,E}^2 \tilde{\sigma}_{\theta_E,G}^2}$. From (41) and proof of Theorem 2, we note that parameters $(\nu_i^\omega, \lambda_i^\omega, \mu_i^\omega, \text{ and } \sigma_i^\omega, i = \{0, 1\})$ are dependent on ω_1 and ω_2 . Thus, we bridge the gap between the allocated weights and detection probability. Based on this mapping relationship, an optimization problem is constructed to seek the optimal weight allocation to achieve maximum detection performance:

$$\max_{\omega_1} P_d^{\omega,\dagger} \quad (54)$$

$$\text{s.t. } \omega_1 + \omega_2 = 1, \quad (54a)$$

$$P_f^\omega = \Gamma. \quad (54b)$$

Since the above optimization problem in (54) is non-convex, conventional convex optimization algorithms are difficult to solve such a problem. Thus, we leverage the SQP method in [66] to obtain the optimal local weights, which has been proven accurate and effective in Section VI.

C. Theoretical Authentication Performance Bound

CRLB is employed to further evaluate the authentication performance of the proposed TTC and LWC schemes under Gaussian AoA. It is a theoretical lower bound for measuring the accuracy of parameter estimation, providing the minimum estimation error variance that an estimator can achieve. In other words, we can utilize such a minimum theoretical estimation error variance to replace the actual one (i.e., $\sigma_{w,\theta}^2$ and $\sigma_{w,c}^2$), achieving the theoretical authentication performance upper bound. CRLB regarding $\sigma_{w,\theta}^2$ and $\sigma_{w,c}^2$ are provided in the following Theorem 4.

Theorem 4: Let $\text{CRB}(\theta)$ and $\text{CRB}(c_t)$ denote the CRLB regarding AoA and MC, respectively. For the received signal in (10), the theoretical lower bounds of $\sigma_{w,\theta}^2$ and $\sigma_{w,c}^2$ are respectively evaluated as

$$\sigma_{w,\theta}^2 \geq \text{CRB}(\theta), \quad (55)$$

$$\sigma_{w,c}^2 \geq \text{CRB}(c_t). \quad (56)$$

Proof: See Appendix D, available online for the detailed calculation of $\text{CRB}(\theta)$ and $\text{CRB}(c_t)$. ■

The derivation of CRLB is of vital importance to establish the fundamental performance limit of our proposed authentication scheme and provides a critical benchmark for evaluation. This rigorous analysis validates the effectiveness of our scheme by measuring its proximity to optimality and revealing its full potential. More detailed results evaluating the gap between the actual performance and this theoretical upper bound are presented in Section VI-H.

TABLE II
SUMMARY OF PARAMETER SETTINGS

Parameter	Value
Carrier Frequency	28 / 60 GHz
Transmit antennas (N_t)	8
The number of snapshots (N)	100
MUSIC grid resolution (δ)	0.05°
MUSIC search range	$[-\pi/2, \pi/2]$
MC coefficients (P)	3
Alice's AoA Mean ($\bar{\theta}_A$)	55°
Eve's AoA Mean ($\bar{\theta}_E$)	52°
Alice's Angular Spread ($\sigma_{\theta_A,\mathcal{L}}$)	0.1887
AoA similarity degree (κ_θ)	4
Alice's MC Mean ($\bar{c}_{t,A}$)	$[1, 0.173 + j0.211, 0.122 + j0.151]^T$
Eve's MC Mean ($\bar{c}_{t,E}$)	$[1, 0.692 + j0.844, 0.488 + j0.604]^T$
Alice's MC Variance ($\sigma_{c,A}^2$)	0.1
Eve's MC Variance ($\sigma_{c,E}^2$)	0.4
MC similarity degree (κ_c)	4
Monte-Carlo trials	3000

VI. PERFORMANCE EVALUATION

A. System Settings

Parameter Settings: We evaluate the proposed synergistic authentication framework in a typical mmWave MIMO scenario. The operating center frequencies are set to 28 GHz and 60 GHz to assess performance under Gaussian and Laplace AoA distributions, respectively. Due to severe signal attenuation, a single LoS path is assumed ($L = 1$). Alice and Eve are positioned at azimuth angles of 55° and 52° , respectively, at a distance of 30 m from Bob. According to the findings in existing channel measurement works [32], [33], the angular spread of Alice at 28 GHz $\sigma_{\theta_A,\mathcal{G}}$ is exponentially distributed with mean $\lambda_{\text{exp}} = 0.1047$ while the angular spread at 60GHz is set as $\sigma_{\theta_A,\mathcal{L}} = 0.1887$. The transceiver utilizes a ULA consisting of $N_t = 8$ antennas. Each transmitted signal frame comprises $N = 100$ snapshots. For feature extraction, the spatial spectrum search is executed over the range $[-\pi/2, \pi/2]$ with a default grid resolution of $\delta = 0.05^\circ$ to ensure high-resolution angular discrimination. The number of MC coefficients is $P = 3$, and the non-zero normalization constant is set as $\xi = 1$. To quantify feature resemblance, the similarity degrees are defined as $\kappa_\theta = \sigma_{\theta_E,\mathcal{G}}/\sigma_{\theta_A,\mathcal{G}}$ ($\sigma_{\theta_E,\mathcal{L}}/\sigma_{\theta_A,\mathcal{L}}$) and $\kappa_c = \sigma_{c,E}^2/\sigma_{c,A}^2$. Performance reliability is validated through 3000 independent Monte-Carlo trials on a 5.4 GHz Intel 14-Core i7 CPU with 32 GB RAM. The comprehensive system configurations and algorithmic parameters are summarized in Table II.

Baseline Settings: To comprehensively evaluate the effectiveness of the proposed scheme, three existing methods are introduced as follows:

- **Hybrid Active-Passive Scheme [22]:** This scheme integrates an active tag embedding feature with passive CG to design a hybrid challenge response authentication mechanism. We adopt an exhaustive search method to find the optimized power allocation ratio between tags and transmitted symbols to maximize the authentication performance.
- **Fine-grained Channel Scheme [28]:** This scheme proposes to combine four physical features including, CG, azimuth AoA, elevation AoA, and PN to achieve device identity

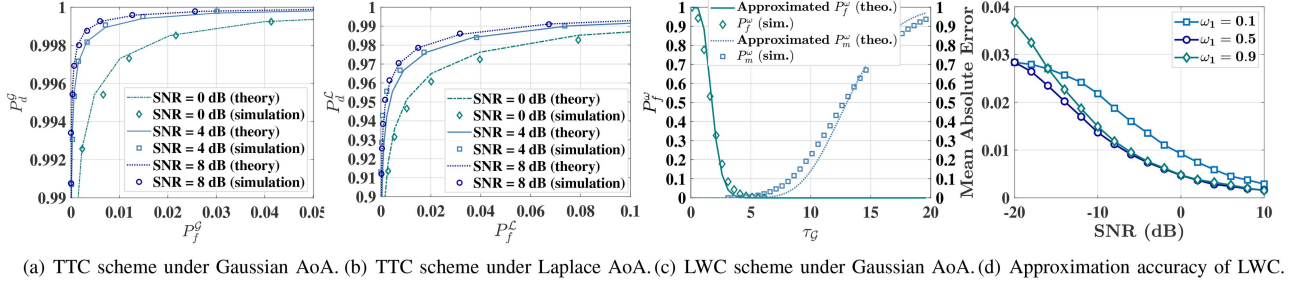


Fig. 5. Model validation under Gaussian and Laplace AoA.

validation. We adopt the identical AoA fingerprint differences as our scheme and conduct the evaluation within a single-path channel environment, ensuring it serves as a standard mmWave channel feature benchmark.

- **Radiation Pattern Error Scheme [38]:** This scheme exploits the radiation pattern features resulting from the antenna array gain, phase, and position errors as device fingerprint and then utilizes the energy detector to realize authentication. We align the statistical discrepancies of three hardware features between legitimate and malicious devices with the MC effects in our scheme under the same antenna configuration.

All evaluated schemes are implemented within a unified 28 GHz mmWave framework with Gaussian-distributed angular features, subjected to identical adaptive spatial spoofing attacks and allocated equivalent pilot overhead. The performance is characterized via receiver operating characteristic curves by sweeping the full range of decision thresholds, thereby ensuring a comprehensive evaluation across all possible false-alarm constraints.

B. Model Validation

To ensure the effectiveness of the theoretical framework under Gaussian and Laplace AoA scenarios, we provide simulation and also theoretical results in Fig. 5(a)–(c) with the settings of $\kappa_\theta = 4$ and $\kappa_c = 4$. For the proposed TTC scheme under considered two conditions, we can observe from Fig. 5(a) and (b) that theoretical curves almost match with simulated ones, implying the developed models can nicely capture the authentication performance in terms of false alarm and detection probabilities under both Gaussian and Laplace AoA. Moreover, we can observe from Fig. 5(c) that the theoretical models in (42) and (43) are close to the simulation results, indicating the effectiveness and feasibility of the derived analytical expressions for the LWC scheme. Another observation reveals that as τ_G increases, P_f^ω decreases while the miss detection probability $P_m^\omega = 1 - P_d^\omega$ presents an increased tendency. This is because a larger τ_G means a wider confidence interval, reducing the occurrence of false alarm events and also making it difficult to detect illegal devices with similar fingerprints.

To further quantify the validity of the sum-weighted approximation, the Mean Absolute Error (MAE) metric is utilized to characterize the discrepancy between the simulated and approximately analytical P_d . As illustrated in Fig. 5(d), the MAE exhibits a monotonic decreasing trend across all three weight configurations as the SNR increases. Specifically, under a fixed low SNR = −20 dB, the MAE are between 0.028 and 0.037. Notably, for SNR values exceeding −10 dB, the MAE consistently

remains below 0.02. This indicates the high reliability of the adopted approximation in most practical authentication scenarios. At SNR = 10dB, the MAE can decrease to less than 0.003, further confirming the effectiveness of the moment-matching approach. In addition, it is noted that precision may degrade under severe noise interference (e.g., SNR < −15 dB). This is primarily because intense noise fluctuations hinder the analytical model from accurately capturing the infinitesimal tail behavior of the non-central components. Nevertheless, even with such degradation, the MAE remains below 0.04, underscoring the practical feasibility of the proposed approximation even in low SNR regimes.

C. Impacts of System Parameters

This subsection investigates in Fig. 6 the effects of several key system parameters on the detection performance P_d of the proposed TTC and LWC methods under Gaussian/Laplace AoA scenarios. For simplicity of expression, we use TTC-G, LWC-G, TTC-L and LWC-L to denote these authentication schemes and utilize P_d to uniformly denote the detection performance of these schemes.

Impact of L on P_d : We first plot in Fig. 6(a) how P_d varies with the number of transmission path $L = \{1, 2, 3\}$ under the settings of ($\kappa_\theta = 4, \kappa_c = 4, \text{SNR} = 0 \text{ dB}$). One can clearly note from Fig. 6(b) that as L increases, P_d tends to increase under four schemes. This is due to the reason that more propagation paths can better reflect the spatial information of the transmitter, providing rich spatial fingerprints and thus improving P_d . It is also noted that even for $L = 1$, all P_d of the proposed four schemes surpass 0.92, implying the proposed two-modal schemes can still effectively detect illegal devices in sparse scattering mmWave systems.

Impact of P on P_d : We proceed to explore the impacts of the number of MC coefficients $P = \{3, 4, 5\}$ on P_d with the settings of ($\kappa_\theta = 4, \kappa_c = 4, \text{SNR} = 0 \text{ dB}$) and summarize the results in Fig. 6(b). As shown in Fig. 6(b), we can see that P_d of all four schemes improves as P increases. Moreover, we observe that P_d of all four schemes surpass 92% even in the scenario of low distinctiveness of MC fingerprint (e.g., $P = 3$). This is because the adopted AoA feature serves as an effective complement and ensures the reliability of the authentication performance. It is also observed from Fig. 6(b) that the LWC-G scheme can further improve the authentication performance and achieve 99% accuracy. The reason is that LWC-G can adaptively assign more weight to the AoA feature to guarantee the robustness of the proposed scheme when the number of MC coefficients is limited.

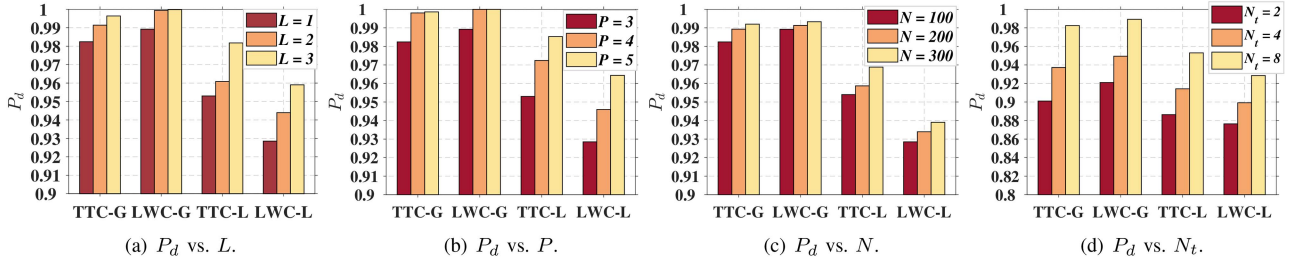


Fig. 6. Impacts of key system parameters on detection performance P_d .

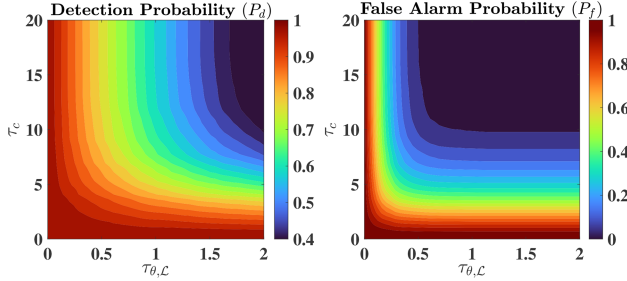


Fig. 7. Sensitivity analysis of the TTC scheme to dual threshold variations.

Impact of N on P_d : To demonstrate the effect of the number of snapshots N on P_d , we illustrate in Fig. 6(c) how P_d varies with N with the settings of ($\kappa_\theta = 4$, $\kappa_c = 4$, SNR = 0 dB). Fig. 6(c) shows that as N becomes larger, P_d under four schemes exhibits an improvement. This is due to the fact that a larger N enables the receiver to extract more accurate physical layer feature parameters and to possess higher device identification ability. In addition, we also notice that the increase in P_d resulting from the number of snapshots is relatively minor compared to the other system parameters like L and P . This is mainly because the improvement of feature extraction accuracy originating from the increased number of snapshots gradually reaches a bottleneck. Nevertheless, the above results also imply that we can obtain satisfied detection performance with fewer snapshots (e.g., $N = 100$) and lower communication overhead.

Impact of N_t on P_d : In Fig. 6(d), we investigate the impact of the number of antenna elements at the transmitter $N_t = \{2, 4, 8\}$ on P_d with the settings of ($\kappa_\theta = 4$, $\kappa_c = 4$, SNR = 0 dB). The number of MC coefficients is set as $P = \{1, 2, 3\}$, respectively. As observed from Fig. 6(d), we note P_d improves as N_t increases under all four schemes. The primary reason is that a larger antenna array provides more interacting antenna elements and thus results in a more unique MC pattern. This increased distinctiveness makes it easier for the receiver to differentiate between legitimate and illegitimate devices. Moreover, a larger transmit array enables higher beamforming gain and improves the SNR at the receiver. This directly contributes to more reliable feature estimation and a higher detection accuracy. One can also see that with a minimal antenna configuration of $N_t = 2$, P_d for all schemes remains above 87%, indicating the effectiveness of our scheme even for UEs equipped with small-scale antenna arrays.

To further demonstrate the sensitivity of the TTC scheme to dual thresholds, we plot in Fig. 7 how the authentication performance varies with different combinations of $\tau_{\theta,L}$ and τ_c values. As observed in the left subplot, P_d exhibits a smooth and continuous degradation rather than a sharp performance drop.

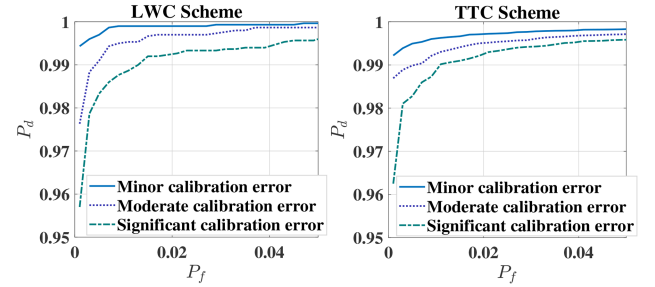


Fig. 8. Robustness of the proposed schemes under different calibration errors.

This indicates that the dual-verification mechanism provides high decision tolerance, that is, a slight calibration deviation in one threshold does not severely degrade the overall detection rate. The right subplot reveals a distinct L-shaped boundary for the false alarm probability performance. The heatmap illustrates that P_f remains stably at a near-zero level across a remarkably broad region of threshold combinations.

D. Robustness Analysis Against Residual Calibration Errors

To assess sensitivity to residual calibration errors, we define three levels of hardware non-ideality: Scenario 1 (Minor errors): $\sigma_g = 0.05$ dB, $\sigma_\phi = 0.5^\circ$, $\sigma_x = 0.005$; Scenario 2 (Moderate errors): $\sigma_g = 0.6$ dB, $\sigma_\phi = 5.5^\circ$, $\sigma_x = 0.08$; and Scenario 3 (Significant errors): $\sigma_g = 0.8$ dB, $\sigma_\phi = 8^\circ$, $\sigma_x = 0.12$. Following the work in [67], the p -th diagonal element of $\tilde{\Gamma}$ is constructed by independently drawing gain and phase residuals from $\mathcal{N}(0, \sigma_g^2)$ and $\mathcal{N}(0, \sigma_\phi^2)$. For the p -th diagonal element of position error matrix $\Delta(\theta)$ (denoted by w_p), we generate the displacements Δx_p from $\mathcal{N}(0, \sigma_x^2)$ to compute $w_p = \exp(j2\pi\Delta x_p \cos \theta / \lambda)$ for the incident path. As illustrated in Fig. 8, both the LWC and TTC schemes exhibit a gradual performance decline as error intensity increases, yet they still maintain satisfactory detection reliability ($P_d > 0.95$ even in a significant error case). The reason is when residual errors are present, the actual steering vector deviates from the assumed ideal model. To minimize the residual error during optimization, the feature extraction algorithm is forced to absorb these unmodeled hardware perturbations into the MC estimates, inevitably leading to biased coefficients. However, we note that these static calibration residuals essentially manifest as intrinsic hardware imperfections of the antenna array. Consequently, the resulting equivalent MC feature essentially enriches the device-specific fingerprint, which potentially compensates for the extraction inaccuracy and ensures robust authentication performance.

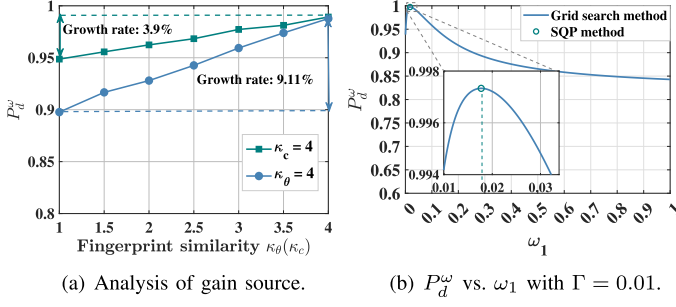


Fig. 9. Optimal feature weight determination of the LWC-G scheme.

E. Optimal Feature Weight Validation of LWC-G Scheme

In this part, we first investigate the contribution of the adopted MC and AoA features to the detection performance. Then, a grid-based exhaustive search method is employed to verify the correctness of the optimized feature weights.

Analysis of Performance Gain Source: We investigate the source of performance gain in two-modal features and summarize results in Fig. 9(a) how P_d^ω varies with the fingerprint similarity κ_θ (resp. κ_c) with $\kappa_c = 4$ (resp. $\kappa_\theta = 4$). Here, we consider $\kappa_\theta = 1$ ($\kappa_c = 1$) to analyze P_d^ω of the proposed scheme with only the MC (or AoA) feature. Note that $\kappa_\theta = 1$ also indicates that the attacker has the same spatial fingerprint as that of a legal device (i.e., simulating theoretical co-located attack scenario). Specifically, we analyze the source of performance gain from two perspectives. On the one hand, the detection accuracy only using the MC feature and the AoA feature are 0.95 and 0.89, respectively. This indicates that the MC feature possesses better device discrimination capability and can exhibit superior P_d^ω even when confronted with co-located attacks. On the other hand, for the fingerprint similarity in the range [1,4], the growth rate of P_d^ω brought by κ_c and κ_θ are 9.11% and 3.9%, respectively. Based on these observations, we can conclude that the MC feature offers higher detection performance gain than the AoA one. Therefore, to achieve better performance in identifying illegal devices, we should assign a higher weight to the MC feature.

Allocated Feature Weight Verification: To validate the effectiveness of the adopted weight determination method based on the optimization problem in (54), we plot in Fig. 9(b) how P_d varies with allocated weight ω_1 under a $\Gamma = 0.01$. In particular, we utilize a step size of 10^{-3} to conduct a grid search for the weight value that maximizes P_d . This grid search-based method is employed to verify the correctness of the optimal weight obtained by the SQP algorithm. The results in Fig. 9(b) indicate that: (1) The obtained weight is consistent with the searched one, verifying the correctness of the designed weight determination framework; and (2) The optimal acquired $\omega_1 = 0.017$ indicates that assigning a smaller weight to AoA and a larger weight to MC can achieve superior detection performance, further verifying that MC contributes more to performance gain. Compared to the high-complexity grid search method, the adopted optimization framework can efficiently decrease the weight calculation time cost (roughly two orders of magnitude faster variations), and thus ensure the effective combination of the considered two-modal features.

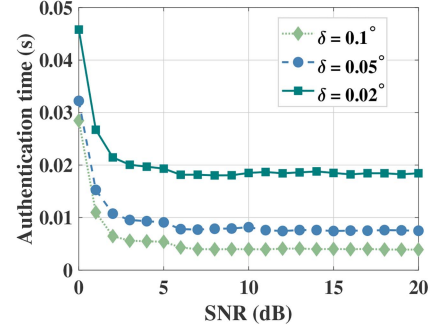


Fig. 10. Authentication time cost.

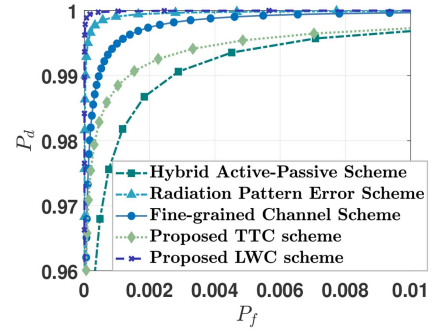


Fig. 11. Performance comparison.

F. Time Cost of Proposed Schemes

In Fig. 10, we present the impact of grid size $\delta = \{0.02^\circ, 0.05^\circ, 0.1^\circ\}$ on the overall authentication time. Given that most time overhead stems from feature extraction, the overall time cost is evaluated by the time spent on feature extraction. We can observe that as δ decreases, the authentication time increases significantly. The reason is that a smaller δ means that there are more candidate points to evaluate in the same search space, leading to more time consumption in the feature extraction phase. Additionally, we also see that the authentication time stabilizes after the SNR reaches 5 dB, primarily due to the enhanced convergence of the MUSIC algorithm to the accurate angle interval at higher signal qualities. Note that even with $\delta = 0.02^\circ$, our scheme maintains an authentication time of approximately 20 ms, which can meet the low-latency identity authentication requirements in mmWave communication systems.

Real-time PHY-layer authentication in mmWave MIMO systems requires balancing accuracy and time cost. In our proposed scheme, this trade-off is primarily governed by the number of snapshots N and the MUSIC grid resolution δ . As shown in Fig. 6(c), increasing the number of snapshots from $N = 100$ to $N = 300$ yields a marginal 2% ~ 4% detection probability gains but linearly triples the covariance matrix estimation complexity ($O(NM^2)$). Similarly, as depicted in Fig. 10, refining δ from 0.1° to 0.02° improves spatial feature extraction but exponentially expands the search space, increasing latency from roughly 5 ms to 20 ms. Based on this quantitative analysis, the configuration of $N = 100$ and $\delta = 0.05^\circ$ emerges as a highly practical operating point. It ensures a robust detection accuracy exceeding 92% while maintaining an overall authentication processing latency of approximately 10 ms. Given that the typical latency budgets in 5G NR and IEEE 802.11ad/ay protocols range from tens to hundreds of milliseconds for initial

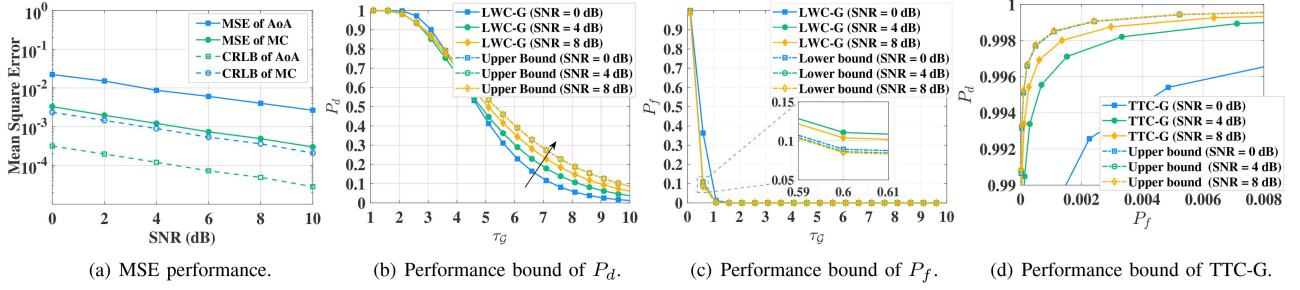


Fig. 12. Theoretical authentication performance bound of LWC-G and TTC-G based on CRLB.

access and security handshakes, this 10 ms overhead proves that our synergistic authentication scheme is viable for real-time deployment.

G. Performance Comparison

To further demonstrate the superiority of the proposed authentication scheme, we conduct a performance comparison with the existing three baselines and summarize the results in Fig. 11. Due to the 28 GHz carrier frequency (i.e., corresponding to Gaussian AoA condition) adopted in [28], we herein only consider the proposed TTC-G and LWC-G methods for fair performance evaluation. As illustrated in Fig. 11, the proposed TTC scheme outperforms the hybrid active-passive scheme but is slightly inferior to the fine-grained channel and radiation pattern error methods. This can be attributed to the fact that while the two-modal synergistic mechanism in TTC provides a substantial security gain over basic hybrid methods, the fine-grained and radiation-pattern schemes leverage more physical features, naturally leading to a higher P_d . However, it is noted that the extraction and processing of such high-dimensional features inevitably impose a significant overhead burden on the mmWave MIMO system.

Another observation from Fig. 11 is that the proposed LWC scheme exhibits the best performance gain across the entire range of P_f compared to the other three baselines. This result indicates a critical advantage: through optimized weight allocation, the LWC method can surpass the performance of higher-dimensional alternatives while utilizing only two-modal features. This is because based on the contribution of each physical layer feature, LWC can reasonably allocate corresponding weight values to fully integrate multiple features and effectively achieve more satisfactory detection performance with lower computational complexity. This makes it more suitable for resource-constrained mmWave communication scenarios.

H. Theoretical Authentication Performance Upper Bound

To evaluate the theoretical performance upper bound of TTC-G and LWC-G, we leverage the derived CRLB in Theorem 4 to replace the variance of feature estimation error (i.e., $\sigma_{w,\theta}^2$ and $\sigma_{w,c}^2$) for obtaining the optimal authentication performance. We provide simulation results regarding feature extraction accuracy in Fig. 12(a), detection performance upper bound of LWC-G in Fig. 12(b), false alarm performance lower bound of LWC-G in Fig. 12(c) and authentication performance upper bound of TTC-G in Fig. 12(d). As shown in Fig. 12(a), it is clear that the MSE of MC is lower than that of AoA for the whole SNR regime and is closer to the corresponding CRLB. This tells us that the

adopted feature extraction method can acquire a more accurate MC fingerprint, indicating that MC can provide higher device identity resolution. From Fig. 12(b) and (c), we observe that the gap between the actual P_d (also P_f) and the theoretical one is gradually smaller as SNR increases. In particular, for a given $\tau_G = 6$, the corresponding gaps in Fig. 12(b) under different SNR are 0.161, 0.1, and 0.045, respectively. From Fig. 12(d), we can obtain similar observations regarding the proposed TTC-G scheme. The above results demonstrate that the developed LWC-G and TTC-G can reach a performance bound under a higher SNR range and exhibit robustness even when confronted with strong interference noise (i.e., SNR = 0 dB).

VII. CONCLUSION

This paper proposed two hybrid PHY-layer authentication schemes exploiting hardware-specific MC and spatial AoA features to achieve device identity validation in a mmWave MIMO system. Moreover, an optimization problem was formulated to determine the optimal feature weights for obtaining improved authentication accuracy. The results in this paper imply that: 1) the proposed schemes exhibit good capability to combat identity impersonation and co-located attacks under both Gaussian and Laplace AoA scenarios; 2) the designed LWC scheme effectively integrates MC and AoA features and fully releases the authentication potential of such adopted features; 3) the inherent antenna array MC effects in mmWave MIMO systems may be exploited to considerably enhance the authentication reliability. Note that the present work is developed under the assumptions of ideal calibration error mitigation and a theoretical framework currently restricted to symmetric zero-mean Laplace distributions. Future PHY-layer authentication research will focus on integrating more calibration residual models to better reflect actual antenna array characteristics. To address the analytical intractability in universal Laplace scenarios, we will explore moment-matching approximations or numerical inversion methods to provide a generalized theoretical performance framework.

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